

The Three-Dimensional Navier–Stokes Regularity Problem

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Abstract

A first singular time would have to end a velocity field that is smooth on every earlier cutoff while leaving no terminal state from which the same Navier–Stokes history can restart. For the maximal classical solution on $[0, T_*)$, we locate the first possible failure in the adjacent trace pair $u \in L^2(0, T_*; H^2)$, $u_t \in L^2(0, T_*; L^2)$. Its cutoff-uniform bound is equivalent to

$$D_* := \int_0^{T_*} \|\Lambda^{3/2} u(t)\|_2^2 dt < \infty,$$

and the critical-height balance makes the same condition equivalent to an upper bound on the signed critical-production prefix. Once D_* is finite, the enstrophy equation supplies a uniform H^1 trace; the Escauriaza–Seregin–Šverák endpoint criterion then excludes a finite T_* . If the corresponding adjacent estimate is available at every finite temporal and spatial order, the rectangles contain identical derivatives on their overlaps. Their compatible intersection reaches one $u_* \in \bigcap_{m < \infty} H^m(\mathbb{R}^3)$, recovers the normalized pressure-complete jet there, and restarts the same history. This proves global smoothness conditional on the whole-terminal rectangle condition.

The argument also identifies the boundary of that condition. Viscosity pays one higher spatial row on every fixed interval $T < T_*$; small critical data absorb the nonlinear production; and any hypothetical finite maximal time obeys the datum-dependent capture horizon $T_* \leq C^4 \|u_0\|_2^4 / (2\nu^5)$. Kinetic energy pays the contribution of every fixed frequency range, so any failure of D_* is ultraviolet. For an arbitrary large Schwartz datum, however, the classical energy and enstrophy balances do not provide the required upper bound on D_* . The manuscript thus gives an exact endpoint reduction and its conditional same-history closure, not an unconditional solution of the three-dimensional regularity problem.

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1 The Regularity Problem

The problem begins with an almost unfair advantage. We may choose the initial fluid as smooth and as rapidly decaying as we like, so there is no hidden roughness waiting in the datum. We then let that field transport itself, make pressure answer incompressibility, and diffuse under viscosity. The question is whether those motions can manufacture a first singular time from perfectly smooth beginnings. Before following that possibility into a stretching vortex, we need to see the fluid that was actually released, the global conclusion it owes us, and the exact place where the classical energy calculation stops paying for that conclusion.

1.1 The equation and the target

Start with the fluid before it has moved. Fefferman's whole-space formulation lets us choose an initial velocity with no roughness to blame later (Fefferman 2000). It is smooth and divergence-free, and every one of its spatial derivatives dies faster than any polynomial as $|x|$ moves out through the fluid. Write this class as

$$\mathcal{S}_\sigma := \left\{ a \in C^\infty(\mathbb{R}^3; \mathbb{R}^3) : \begin{array}{l} \nabla \cdot a = 0, \\ \forall \alpha \in \mathbb{N}_0^3 \forall K \in \mathbb{N}_0 \exists C_{\alpha,K} < \infty \forall x \in \mathbb{R}^3, \\ |\partial_x^\alpha a(x)| \leq C_{\alpha,K} (1 + |x|)^{-K} \end{array} \right\}. \quad (1.1)$$

The quantifiers matter. Choose a derivative α , then choose any algebraic decay rate K , however severe. A finite constant $C_{\alpha,K}$ still controls that derivative everywhere in space. The constant may change when either choice changes, but it does not drift with x . This is the divergence-free Schwartz class.

For a finite spatial height m , let

$$H_\sigma^m(\mathbb{R}^3) := \{v \in H^m(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0\}.$$

The decay in (1.1) places the same datum in every one of these spaces. To see it, fix $K > 3/2$. For $|\alpha| \leq m$,

$$\|\partial_x^\alpha a\|_2^2 \leq C_{\alpha,K}^2 \int_{\mathbb{R}^3} (1 + |x|)^{-2K} dx < \infty.$$

Only finitely many derivatives occur below a chosen finite m , and hence

$$\mathcal{S}_\sigma \subset H_\sigma^m(\mathbb{R}^3) \quad \text{for every finite } m. \quad (1.2)$$

At time zero, every finite Sobolev height is already present. If one of those heights later escapes, the evolution itself has created the loss.

Fix one $u_0 \in \mathcal{S}_\sigma$ and one viscosity $\nu > 0$. The velocity u and the pressure p obey one coupled law:

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(\cdot, 0) = u_0. \quad (1.3)$$

Read this as one event. The time derivative records how the field changes. Transport lets the velocity carry itself through the motion it has already created. Incompressibility does not permit that motion to squeeze volume, so pressure answers across the surrounding field. Viscosity acts on the curvature of the resulting velocity. Every later derivative in the paper is another reading of this coupled law.

Local theory starts the motion smoothly, and uniqueness joins its local pieces into one maximal classical history on $[0, T_*)$, where $T_* \in (0, \infty]$. The Millennium Prize problem asks whether this interval can have a finite right edge. In the notation above, the desired conclusion is

$$T_* = \infty, \quad u, p \in C^\infty(\mathbb{R}^3 \times [0, \infty)), \quad \sup_{t \geq 0} \int_{\mathbb{R}^3} |u(x, t)|^2 dx < \infty. \quad (1.4)$$

The last slice is the issue. A field may be smooth on every shorter interval $[0, T]$, $T < T_*$, while a derivative loses every finite trace as $t \uparrow T_*$. Smoothness before the endpoint and arrival at the endpoint are different facts. A global argument has to carry enough of the velocity to that face to restart the equation, or make the face impossible.

The first global calculation asks what the equation can already preserve. Pair (1.3) with u across the whole fluid:

$$\frac{1}{2} \frac{d}{dt} \|u(t)\|_2^2 + \int_{\mathbb{R}^3} (u \cdot \nabla) u \cdot u dx + \int_{\mathbb{R}^3} \nabla p \cdot u dx = \nu \int_{\mathbb{R}^3} \Delta u \cdot u dx.$$

Transport can make the velocity steeper locally, but incompressibility turns its contribution here into a flux through space:

$$\int_{\mathbb{R}^3} (u \cdot \nabla) u \cdot u dx = \frac{1}{2} \int_{\mathbb{R}^3} u \cdot \nabla |u|^2 dx = -\frac{1}{2} \int_{\mathbb{R}^3} (\nabla \cdot u) |u|^2 dx = 0.$$

Pressure meets the same constraint,

$$\int_{\mathbb{R}^3} \nabla p \cdot u dx = - \int_{\mathbb{R}^3} p \nabla \cdot u dx = 0,$$

while viscosity is left with a fixed sign:

$$\nu \int_{\mathbb{R}^3} \Delta u \cdot u dx = -\nu \|\nabla u\|_2^2.$$

After integrating in time, the balance reads

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 ds = \frac{1}{2} \|u_0\|_2^2. \quad (1.5)$$

The datum now controls the amount of velocity at every time and the accumulated square of its gradient. It does not control that gradient at each time. A steeper change can occupy a shorter interval, become steeper again on a still shorter interval, and leave the integral in (1.5) finite through the entire sequence.

That is the opening. The fluid can keep a calm kinetic-energy account while its motion is pressed across a narrowing distance. In three dimensions, vortex stretching is the part of the equation capable of attempting exactly that. We now follow one such vortex as it narrows, find the scale at which energy stops seeing the danger, and ask what the velocity must do in time if the narrowing is actually to continue.

2 Critical Height and the Derivative Tower

The energy balance left one precise freedom. It controls the amount of velocity and the time accumulated by one spatial derivative, but it does not stop the same velocity change from being pressed through successively smaller distances. In three dimensions the equation contains a mechanism capable of attempting that compression: a vortex may be stretched along its axis while incompressibility contracts its cross-section. We follow the same vortex through that motion. Its narrowing forces us to compare stretching with viscosity, the comparison exposes the scale the energy balance cannot see, and a finite terminal time then forces the fluid to cross its changing states with less and less time left. Velocity, acceleration, jerk, and the rest of the derivative Tower enter because the proposed singular history must produce those changes; they are not machinery added from outside the equation.

2.1 The critical-height problem inside a stretching vortex

Follow the maximal history into a region where the fluid is rotating coherently and the surrounding velocity field is pulling along the direction of that rotation. The two local objects are

$$\omega := \nabla \times u, \quad S := \frac{1}{2}(\nabla u + (\nabla u)^T).$$

Taking the curl removes the pressure gradient but not the motion through which pressure helped create the velocity field. The rotating part of that same field obeys

$$\partial_t \omega + (u \cdot \nabla) \omega = S \omega + \nu \Delta \omega. \quad (2.1.1)$$

The right-hand side contains the entire contest. Viscosity diffuses variation in the vorticity. The term $S \omega$ is the action of the local strain on the direction and magnitude of the rotation. When ω points along an extensional direction of S ,

$$\omega \cdot S \omega > 0,$$

and the surrounding motion feeds the rotation rather than flattening it.

It cannot receive that amplification while keeping the same shape. Incompressibility says

$$\text{tr } S = \nabla \cdot u = 0.$$

so extension along the vortex axis must be paid for by contraction across the axis. The same rotating structure becomes longer and thinner. A velocity variation ΔU that once occupied a transverse width ℓ is now carried through less space, and its gradient has the scale

$$|\nabla u| \sim \frac{\Delta U}{\ell}.$$

The numerator need not become infinite for the ratio to do so. A finite velocity difference can become an unbounded derivative if the distance across which the fluid makes that difference is allowed to collapse.

Nothing in the kinetic-energy balance forbids this geometry. For the same solution, that balance is

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 ds = \frac{1}{2} \|u_0\|_2^2. \quad (2.1.2)$$

The first term measures velocity across the volume of the fluid. It does not remember the width across which any one part of that velocity changes. The second term pays for the square of the gradient only after time has been integrated. A steep event may occupy a short interval, a steeper event a shorter one, and the account may remain finite while the pointwise spatial profile becomes harder and harder to resolve.

The narrowing also strengthens viscosity, so stretching by itself has not identified a route to singularity. Pair (2.1.1) with ω . Before integrating through space, the rotation satisfies

$$\frac{1}{2}(\partial_t + u \cdot \nabla)|\omega|^2 = \omega \cdot S\omega + \frac{\nu}{2}\Delta|\omega|^2 - \nu|\nabla\omega|^2.$$

After integration over $\mathbb{R}^3$, transport carries enstrophy through the fluid without changing its total amount, and the Laplacian flux vanishes at spatial infinity. What remains is the exact competition

$$\frac{1}{2} \frac{d}{dt} \|\omega(t)\|_2^2 + \nu \|\nabla\omega(t)\|_2^2 = \int_{\mathbb{R}^3} \omega \cdot S\omega dx. \quad (2.1.3)$$

The term on the right can produce enstrophy; the term on the left removes it. Both act on the same rotating structure and both become stronger as its width falls. Their magnitudes at one chosen width cannot tell us which action gains when the whole structure is made smaller. For that we have to shrink the law itself and watch how every term changes together.

So shrink the entire equation without changing the law that governs it. For $\lambda > 0$, define

$$u_\lambda(x, t) := \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) := \lambda^2 p(\lambda x, \lambda^2 t). \quad (2.1.4)$$

The time response, transport, pressure response, and viscous response all receive the same factor λ^3 :

$$\begin{aligned} \partial_t u_\lambda &= \lambda^3 (\partial_t u)(\lambda x, \lambda^2 t), & (u_\lambda \cdot \nabla) u_\lambda &= \lambda^3 ((u \cdot \nabla) u)(\lambda x, \lambda^2 t), \\ \nabla p_\lambda &= \lambda^3 (\nabla p)(\lambda x, \lambda^2 t), & \Delta u_\lambda &= \lambda^3 (\Delta u)(\lambda x, \lambda^2 t). \end{aligned}$$

The smaller copy has not handed viscosity an extra power with which to defeat transport. The terms remain locked in the proportion already fixed by the equation. What has changed is the room and the time: space contracts by λ^{-1} , and the same lawful motion is run on a clock shorter by λ^{-2} .

We now need the spatial reading that does not fade when the structure shrinks and does not inflate simply because we have magnified it. Let $\Lambda = (-\Delta)^{1/2}$. At Sobolev height s , a change of variables gives

$$\|u_\lambda(t)\|_{\dot{H}^s}^2 = \lambda^{2s-1} \|u(\lambda^2 t)\|_{\dot{H}^s}^2. \quad (2.1.5)$$

There is one height at which the scale factor disappears:

$$2s - 1 = 0, \quad s = \frac{1}{2}.$$

This selects

$$H(t) := \frac{1}{2} \|u(t)\|_{\dot{H}^{1/2}}^2 = \frac{1}{2} \|\Lambda^{1/2} u(t)\|_2^2 \quad (2.1.6)$$

as the critical height. A lawful concentration cannot lower this reading by making itself smaller. Kinetic energy sits below it and loses one power of λ ; the first derivative sits above it and gains one:

$$\|u_\lambda(t)\|_2^2 = \lambda^{-1} \|u(\lambda^2 t)\|_2^2, \quad H_\lambda(t) = H(\lambda^2 t), \quad \|\nabla u_\lambda(t)\|_2^2 = \lambda \|\nabla u(\lambda^2 t)\|_2^2.$$

The same contraction can now make energy quieter and a derivative louder. Between them, the critical height keeps the size of the concentration fixed. It is the scale at which the narrowing can no longer hide behind the energy balance.

One rescaled profile lets us watch all three readings at once. For a nonzero divergence-free $\phi \in \mathcal{S}_\sigma$, set

$$u_\ell(x) := \ell^{-1} \phi(x/\ell).$$

Then

$$\begin{aligned} \|u_\ell\|_\infty &\sim \ell^{-1}, \quad \|\nabla u_\ell\|_\infty \sim \ell^{-2}, \\ \|u_\ell\|_2^2 &\sim \ell, \quad \|u_\ell\|_{\dot{H}^{1/2}}^2 = \|\phi\|_{\dot{H}^{1/2}}^2. \end{aligned}$$

As $\ell \downarrow 0$, the amplitude and gradient rise, the energy falls, and the critical height does not move. These profiles are not successive states of a Navier–Stokes solution; nothing here says that the evolution can produce them. They isolate the shape of the loophole: energy allows it, scaling preserves it, and the nonlinear equation still has to decide whether the actual fluid can travel through it.

Scaling has selected the height at which a shrinking structure remains fully visible. We can now return to the equation and ask how the actual solution moves through that height. For

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2,$$

keep the transport and pressure generated by the same velocity together in the signed response

$$\mathcal{P}_s(t) := -\operatorname{Re} \left\langle \Lambda^s u, \Lambda^s \left((u \cdot \nabla) u + \nabla p \right) \right\rangle_{L^2}.$$

At this global scalar level the pressure contribution vanishes after incompressibility is used. It remains in the definition because the balance was produced by the pressure-complete momentum equation, and the differentiated rows later in the paper must remember that origin. Pairing at height s gives

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}. \quad (2.1.7)$$

At the critical height,

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2}. \quad (2.1.8)$$

Thus a positive nonlinear production can raise H only against the viscous payment $2\nu E_{3/2}$ made at the same instant. The geometric contest in the vortex has become one exact scalar row. Neither stretching nor diffusion can win this row by changing scale; the unresolved issue

is whether the signed production can keep outrunning its adjacent viscous cost along the actual history.

Now suppose, only to follow the possible failure, that the maximal history has a finite endpoint T_* . The endpoint criterion of Escauriaza, Seregin, and Šverák says that a uniform L^3 ceiling would continue the solution through that endpoint (Escauriaza, G. A. Seregin, and Šverák 2003). The critical Sobolev embedding reads

$$\|u(t)\|_3 \leq C \left\| \Lambda^{1/2} u(t) \right\|_2 = C \sqrt{2H(t)}. \quad (2.1.9)$$

Consequently the same hypothetical endpoint forces

$$\sup_{0 \leq t < T_*} H(t) = \infty. \quad (2.1.10)$$

This is no longer a sequence of profiles chosen outside the evolution. The actual velocity history must climb through arbitrarily large critical levels. If

$$M_n := 2^n \max\{1, H(0)\}, \quad t_n := \inf\{t < T_* : H(t) = M_n\},$$

then continuity gives $t_n \uparrow T_*$. The first-passage intervals are successive pieces of one history, and

$$\sum_{n=0}^{\infty} (t_{n+1} - t_n) = T_* - t_0 < \infty. \quad (2.1.11)$$

The heat equation now gives this finite-time ascent its spatial clock. If viscosity is acting on a structure of width r , its linear comparison time is determined without analogy. For $v_t = \nu \Delta v$,

$$\hat{v}(\xi, t + \delta t) = e^{-\nu|\xi|^2 \delta t} \hat{v}(\xi, t).$$

A structure of width r lives near $|\xi| \simeq r^{-1}$, so diffusion changes its amplitude by order one on the time

$$\tau_\nu(r) \asymp \frac{r^2}{\nu}. \quad (2.1.12)$$

This does not say that the nonlinear fluid forms that width, or that one width can be read uniquely from H . It says what time viscosity assigns to a scale once the evolution has formed it.

Take geometrically shrinking widths

$$r_n := 2^{-n} r_0, \quad \tau_n := \frac{r_n^2}{\nu}.$$

Their heat clocks satisfy

$$\tau_n = \frac{r_0^2}{\nu} 4^{-n}, \quad \sum_{n=0}^{\infty} \tau_n = \frac{4r_0^2}{3\nu} < \infty. \quad (2.1.13)$$

Thus the viscous clocks attached to infinitely many geometrically smaller scales fit inside a finite amount of elapsed time. The arithmetic agrees with the actual first-passage compression in (2.1.11); it does not identify those passages with the model widths r_n .

For a singularity to use this opening, the fluid would have to form r_0 , then r_1 , then r_2 , without losing the stretching alignment while viscosity acts on every scale already formed. Each event would begin from the velocity produced by the previous event. The widths would shrink, the clocks would shorten, and the critical production in (2.1.8) would have to meet its viscous cost again and again.

The spatial picture has now become a temporal obligation. The same velocity must cross unbounded critical levels during intervals whose total length is finite. Those changes have to be carried by its time response; if that response also escapes, its changes have to be carried by the next one. This is where the temporal derivative Tower begins.

2.2 Velocity, acceleration, jerk, and the Tower

The finite endpoint has already forced the same velocity history through unbounded critical levels. Naming those levels does not move the fluid between them. On every strict subinterval that movement is carried by the fundamental theorem of calculus. Let X be one fixed spatial Banach space in which the classical responses below are defined. For every $n \geq 0$,

$$\|\partial_t^n u(t_1) - \partial_t^n u(t_0)\|_X \leq \int_{t_0}^{t_1} \|\partial_t^{n+1} u(s)\|_X ds. \quad (2.2.1)$$

If $\partial_t^{n+1} u$ had a uniform roof before a finite T_* , its integral would give $\partial_t^n u$ a uniform roof as well. Reading the contrapositive upward, under the fixed-space and absolute-continuity hypotheses spelled out in Section B.2,

$$\sup_{0 < t < T_*} \|u(t)\|_X = \infty \implies \sup_{0 < t < T_*} \|\partial_t^n u(t)\|_X = \infty \quad \text{for every finite } n. \quad (2.2.2)$$

The critical escape in (2.1.10) supplies the premise with $X = \dot{H}^{1/2}(\mathbb{R}^3)$. It does not prove that a finite endpoint exists. It says that if the history ends there, no finite Eulerian time row may remain uniformly calm while the lower row escapes:

$$u, \quad \partial_t u, \quad \partial_t^2 u, \quad \partial_t^3 u, \quad \dots$$

Write

$$V_n := \partial_t^n u, \quad Q_n := \partial_t^n p, \quad n \in \mathbb{N}_0. \quad (2.2.3)$$

These are the velocity, temporal acceleration, temporal jerk, and higher Eulerian responses of the fluid. They are read at a fixed spatial position. The acceleration of a particle moving with the fluid is instead

$$D_t u = \partial_t u + (u \cdot \nabla) u.$$

The transport term has not been discarded; the Eulerian Tower keeps it inside the recurrence. The Tower is not an extra model placed beside Navier–Stokes. The original equation generates every rung.

The first row is simply the original equation solved for its time response:

$$V_1 = \nu \Delta V_0 - (V_0 \cdot \nabla) V_0 - \nabla Q_0.$$

When time is differentiated once, the derivative may land on either factor in the transport term:

$$V_2 = \nu \Delta V_1 - (V_1 \cdot \nabla) V_0 - (V_0 \cdot \nabla) V_1 - \nabla Q_1.$$

At the next step, there are two ways to place one derivative on each transport factor, and the middle term records both:

$$V_3 = \nu \Delta V_2 - (V_2 \cdot \nabla) V_0 - 2(V_1 \cdot \nabla) V_1 - (V_0 \cdot \nabla) V_2 - \nabla Q_2.$$

The coefficients are not decoration. They count every way the n time derivatives can be distributed across the velocity doing the transporting and the velocity being transported. The complete row is

$$V_{n+1} + \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} + \nabla Q_n = \nu \Delta V_n, \quad \nabla \cdot V_n = 0. \quad (2.2.4)$$

Pressure does not get to evolve independently beside these rows. Incompressibility determines it from the velocity at the same time. Taking the divergence of the base equation and choosing the decaying whole-space normalization gives

$$-\Delta p = \partial_i \partial_j (u_i u_j), \quad p = R_i R_j (u_i u_j), \quad (2.2.5)$$

where $R_j = \partial_j (-\Delta)^{-1/2}$. After n time derivatives,

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j. \quad (2.2.6)$$

Thus the velocity rows determine Q_n , and ∇Q_n returns to (2.2.4) as part of the very response those rows generate. Transport, pressure, incompressibility, and viscosity remain one coupled event at every height of the Tower.

The temporal climb has now exposed a spatial pressure. Every new response contains ΔV_n ; transport differentiates products of the earlier responses; pressure is recovered from spatial derivatives of those products. We entered the Tower because the hypothetical endpoint made the fluid respond faster and faster. The equation has answered by placing higher spatial order inside every new response. We return to the critical height to see exactly what that spatial direction contains before naming the full two-index object.

2.3 The Tower opens toward more spatial regularity

The proposed failure drove the vortex toward smaller scales because spatial control was supposed to disappear there. Finite time then drove the same history upward through its temporal responses. But the equation that generates each response contains $\nu \Delta$ of the response below it. The attempt to lose spatial regularity has opened an algebraic direction toward more of it. We now have to find out what that direction actually pays for.

The critical height puts the temporal and spatial movements on the same scale. Under (2.1.4),

$$H_\lambda(t) = H(\lambda^2 t), \quad H_\lambda^{(n)}(t) = \lambda^{2n} H^{(n)}(\lambda^2 t).$$

The spatial energy at height $n + \frac{1}{2}$ carries exactly that weight:

$$E_{n+1/2,\lambda}(t) = \lambda^{2n} E_{n+1/2}(\lambda^2 t).$$

The n -th time response of H and the spatial energy at height $n + \frac{1}{2}$ carry the same scale weight. This coincidence does not yet relate their values. The critical balance does.

Return to the exact balance already found in (2.1.7):

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}.$$

At critical height,

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2}.$$

Differentiate once and use the same balance at the next spatial height:

$$\begin{aligned} H'' &= \partial_t \mathcal{P}_{1/2} - 2\nu E'_{3/2} \\ &= \partial_t \mathcal{P}_{1/2} - 2\nu \mathcal{P}_{3/2} + (2\nu)^2 E_{5/2}. \end{aligned}$$

One more derivative gives

$$H''' = \partial_t^2 \mathcal{P}_{1/2} - 2\nu \partial_t \mathcal{P}_{3/2} + (2\nu)^2 \mathcal{P}_{5/2} - (2\nu)^3 E_{7/2}.$$

The new top spatial energy has appeared, but nothing below it has disappeared. Every differentiated transport–pressure response remains in the row. The pattern continues, and for $n \geq 1$, induction gives

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}. \quad (2.3.1)$$

Because $\nu > 0$, the completed row can be solved for its top spatial energy:

$$E_{n+1/2} = \frac{H^{(n)} - \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}}{(-2\nu)^n}. \quad (2.3.2)$$

This algebra exposes the higher energy; it does not bound it. Every transfer term on the right remains, and controlling the new energy requires controlling that entire lower triangular chain. What has changed is the direction of the problem. Higher temporal response places

$$E_{1/2}, \quad E_{3/2}, \quad E_{5/2}, \quad E_{7/2}, \quad \dots$$

inside the completed critical-height Tower rather than carrying the argument downward toward weaker spatial information.

The scalar rows still have to be tied back to the velocity that produced them. Because

$$H(t) = \frac{1}{2} \left\langle \Lambda^{1/2} V_0(t), \Lambda^{1/2} V_0(t) \right\rangle_{L^2},$$

Leibniz's rule gives

$$H^{(n)}(t) = \frac{1}{2} \sum_{a=0}^n \binom{n}{a} \operatorname{Re} \left\langle \Lambda^{1/2} V_a(t), \Lambda^{1/2} V_{n-a}(t) \right\rangle_{L^2}. \quad (2.3.3)$$

Thus $H^{(n)}$ is neither the response V_n nor the spatial energy $E_{n+1/2}$. It is a quadratic contraction of the temporal responses, and (2.3.2) can expose the top spatial energy only after the complete transport–pressure row has been carried with it.

Only now do we have two independent directions that need one notation. For a temporal height n and a spatial multi-index α , set

$$J_{n,\alpha} := \partial_t^n \partial_x^\alpha u, \quad \Pi_{n,\alpha} := \partial_t^n \partial_x^\alpha p, \quad \alpha \in \mathbb{N}_0^3. \quad (2.3.4)$$

Differentiating the same momentum equation in both directions gives

$$\begin{aligned} J_{n+1,\alpha} + \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta} + \nabla \Pi_{n,\alpha} \\ = \nu \Delta J_{n,\alpha}, \quad \nabla \cdot J_{n,\alpha} = 0. \end{aligned} \quad (2.3.5)$$

The pressure coordinate is still slaved to those velocity coordinates:

$$-\Delta \Pi_{n,\alpha} = \partial_x^\alpha \partial_i \partial_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j. \quad (2.3.6)$$

The indices have not split the fluid into separate models. They tell us which time response of the same velocity is being read, and how finely that response is being resolved in space.

Fix one temporal response V_r . Pairing its differentiated equation at a spatial height s repeats the same adjacent movement. Set

$$E_{r,s} := \frac{1}{2} \|\Lambda^s V_r\|_2^2$$

and let $\mathcal{P}_{r,s}$ be the complete pairing of V_r with the transport and pressure terms in its differentiated equation. Then

$$E'_{r,s} = \mathcal{P}_{r,s} - 2\nu E_{r,s+1}. \quad (2.3.7)$$

The equation now names the spatial column that belongs to this response:

$$V_r \in H_x^0, \quad V_r \in H_x^1, \quad V_r \in H_x^2, \quad V_r \in H_x^3, \quad \dots$$

Temporal order chooses which response of the fluid is being read; spatial order chooses how finely that response is resolved. These membership statements are the coordinates the proof has to establish, not consequences of (2.3.7) by itself. The nonlinear pairings in $\mathcal{P}_{r,s}$ still

have to be bounded, first at a finite collection of adjacent heights and eventually uniformly as the terminal time is approached.

This tells us why an intersection will eventually enter, but the order matters. For a fixed finite r and m , one must first prove that the corresponding coordinate remains finite on the interval under consideration. If that can be done for every finite spatial height of one response, then

$$\bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

If the same payment can then be made for every finite temporal response, the common object is

$$\bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I; H^m(\mathbb{R}^3)). \quad (2.3.8)$$

Every prescribed temporal derivative and spatial derivative uses only a finite coordinate, followed by the corresponding Sobolev embedding. The infinite intersection says that all of those finite payments belong to one compatible history. It cannot be inferred from the diagram alone.

We have reached the reversal without pretending to have completed it. A finite singular history must ascend through faster temporal responses while losing spatial control. Viscosity places an adjacent higher spatial energy in every one of those response rows. The remaining problem is to turn that algebraic adjacency into estimates on finite pieces of the actual history and then ask whether those estimates survive the terminal limit. That finite payment is what the two-dimensional rectangles are built to measure.

3 The Intersection Inside the Tower

The Tower has left us with a distinction the rest of the argument cannot blur. Viscosity exposes a higher spatial energy in every temporal row, but the nonlinear transfers remain beside it, so exposure is not control. We now find the place where viscosity does make an honest spatial payment: first on a fixed interval strictly before T_* , then in the adjacent pair of rows that can reach a time slice. The trace calculation in Theorem C.3.1 turns that pair into the smallest finite rectangle. Once the cutoff is allowed to approach T_* , the entire regularity problem is concentrated in whether the same rectangle keeps one finite bound.

3.1 A spatial column inside every temporal response

Before estimating anything, fix the algebraic point at which the two directions meet. Begin with $V_0 = u$, whose critical height first exposed the spatial movement. For $s \geq 0$, set

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2$$

and retain the complete transport–pressure pairing

$$\mathcal{P}_s(t) := -\operatorname{Re} \left\langle \Lambda^s u, \Lambda^s \left((u \cdot \nabla) u + \nabla p \right) \right\rangle_{L^2}.$$

The momentum equation paired at that height is

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}. \quad (3.1.1)$$

At $s = \frac{1}{2}$,

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2}. \quad (3.1.2)$$

Differentiate this same row. The first two steps are

$$H'' = \partial_t \mathcal{P}_{1/2} - 2\nu \mathcal{P}_{3/2} + (2\nu)^2 E_{5/2}, \quad (3.1.3)$$

$$H''' = \partial_t^2 \mathcal{P}_{1/2} - 2\nu \partial_t \mathcal{P}_{3/2} + (2\nu)^2 \mathcal{P}_{5/2} - (2\nu)^3 E_{7/2}. \quad (3.1.4)$$

Each differentiation exposes one higher spatial energy, and each lower transport–pressure transfer survives beside it.

At finite order n , the complete identity is

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}, \quad n \geq 1. \quad (3.1.5)$$

Its induction from (3.1.1) is carried out in Proposition B.4.1. Because $\nu > 0$, the row can be solved for its top coordinate:

$$E_{n+1/2} = \frac{H^{(n)} - \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}}{(-2\nu)^n}. \quad (3.1.6)$$

The division has not erased the transfer sum. It has isolated the exact collection that must be controlled before the exposed spatial energy is controlled. This is where the hoped-for reversal becomes conditional: viscosity has put a higher energy in the equation, but writing the equation has not bounded the nonlinear chain on its right.

The height derivative remains a contraction of the velocity responses:

$$H^{(n)} = \frac{1}{2} \sum_{a=0}^n \binom{n}{a} \operatorname{Re} \langle \Lambda^{1/2} V_a, \Lambda^{1/2} V_{n-a} \rangle_{L^2}. \quad (3.1.7)$$

The three readings at order n must remain distinct. V_n is the Eulerian response of the velocity, $H^{(n)}$ is a quadratic contraction of the temporal responses, and $E_{n+1/2}$ is the highest spatial energy exposed after the full transfer chain has been retained.

Table 3.1.1: The temporal response and the spatial height exposed at the same order.

order	velocity response	critical response	exposed height
0	$V_0 = u$	$H = E_{1/2}$	$E_{1/2}$
1	$V_1 = u_t$	H'	$E_{3/2}$
2	$V_2 = u_{tt}$	H''	$E_{5/2}$
3	$V_3 = u_{ttt}$	H'''	$E_{7/2}$
n	$V_n = \partial_t^n u$	$H^{(n)}$	$E_{n+1/2}$

Now shift the base from u to any fixed response V_r . The Laplacian still acts linearly on that same response, and every split of the transport derivative remains in the binomial sum. The question is no longer whether a higher spatial coordinate appears. It is where the equation actually pays for that coordinate on the history already in hand.

3.2 Where viscosity pays on a strict interval

Fix one response V_r and one spatial height s . Its scalar energy and complete transfer are

$$E_{r,s} := \frac{1}{2} \|\Lambda^s V_r\|_2^2$$

and

$$\mathcal{P}_{r,s} := -\operatorname{Re} \left\langle \Lambda^s V_r, \Lambda^s \left[\sum_{a=0}^r \binom{r}{a} (V_a \cdot \nabla) V_{r-a} + \nabla Q_r \right] \right\rangle_{L^2}.$$

Pairing the r -th response equation at this height gives

$$E'_{r,s} = \mathcal{P}_{r,s} - 2\nu E_{r,s+1}. \quad (3.2.1)$$

Beginning at $s = \frac{1}{2}$ and repeating the induction in Proposition B.4.1 produces

$$\partial_t^n E_{r,1/2} = (-2\nu)^n E_{r,n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{r,j+1/2}. \quad (3.2.2)$$

The pattern repeats inside every response, together with the same obstruction: the higher spatial energy appears, and the nonlinear transfer remains beside it.

To see an actual spatial gain, stop first at one strict cutoff $T < T_*$ and write $I = (0, T)$. Apply the Leray projection to the differentiated momentum equation. It removes the pressure gradient from this estimate while leaving the pressure-complete recurrence intact:

$$\partial_t V_r - \nu \Delta V_r = -F_r, \quad F_r := \mathbb{P} \sum_{a=0}^r \binom{r}{a} (V_a \cdot \nabla) V_{r-a}.$$

For an integer $m \geq 2$, the finite-order product bound proved in Proposition A.1.1 gives

$$\|F_r\|_{H^{m-1}} \leq C_{m,r} \sum_{a=0}^r \|V_a\|_{H^m} \|V_{r-a}\|_{H^m}. \quad (3.2.3)$$

The same local construction preserves every finite derivative already present in the smooth datum on this chosen interval. For the finite indices now in use,

$$\max_{0 \leq a \leq r} \|V_a\|_{L^\infty(0,T;H^m)} < \infty. \quad (3.2.4)$$

Hence the forcing in (3.2.3) is square-integrable in time.

Pair the projected equation one spatial step above its forcing. The heat term now makes the payment that the triangular identity only exposed:

$$\frac{d}{dt} \|V_r\|_{H^m}^2 + \nu \|V_r\|_{H^{m+1}}^2 \leq C_{\nu,m} (\|V_r\|_{H^m}^2 + \|F_r\|_{H^{m-1}}^2). \quad (3.2.5)$$

After time integration, V_r gains one spatial degree. Returning to the equation puts its next time response one degree below:

$$V_r \in L^2(I; H^{m+1}), \quad V_{r+1} = \partial_t V_r \in L^2(I; H^{m-1}). \quad (3.2.6)$$

This is the first place where the spatial movement has been estimated rather than read from an identity.

The finite estimates may be placed beside one another as a two-dimensional grid. Moving down changes the temporal response; moving across changes the spatial resolution of that response.

Table 3.2.1: The mixed derivative grid. Each dot names a coordinate of the same velocity; no dot is a cutoff-uniform endpoint estimate.

	H_x^0	H_x^1	H_x^2	H_x^3	$\dots$
$V_0 = u$	•	•	•	•	$\dots$
$V_1 = u_t$	•	•	•	•	$\dots$
$V_2 = u_{tt}$	•	•	•	•	$\dots$
$V_3 = u_{ttt}$	•	•	•	•	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Now ask for a finite observation of the solution: q time derivatives and k spatial derivatives. Choose finite heights

$$R > q + \frac{1}{2}, \quad M > k + \frac{3}{2}.$$

The vector-valued time embedding and the three-dimensional spatial embedding (Cerdà 2010, Theorem 7.28) give

$$H^R(I; H^M(\mathbb{R}^3)) \hookrightarrow C^q(\bar{I}; C^k(\mathbb{R}^3)).$$

Repeating the finite payment high enough in each direction fills every finite request on this fixed I . Collecting those requests on the same history gives

$$\mathcal{V}_\infty(I) := \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I; H^m(\mathbb{R}^3)) \hookrightarrow C^\infty(\bar{I} \times \mathbb{R}^3). \quad (3.2.7)$$

The constants may depend on the chosen finite coordinate. Smoothness asks for each such coordinate in turn, not for one constant uniform over the whole grid.

The cutoff is the unpaid part. Every dot is finite after $T < T_*$ has been fixed, but none of the estimates above prevents its constant from diverging as $T \uparrow T_*$. To carry one spatial coordinate to that time slice, it must be paired with the temporal response directly below it. Those adjacent rows form the finite rectangles.

3.3 The finite piece that reaches a time slice

Let $I_* = (0, T_*)$ be the interval of the proposed finite maximal history, and stop first at a strict cutoff $T < T_*$. Control of a response one spatial step above H^M does not by itself place that response continuously at $t = T$. Its time derivative must approach the same trace from one spatial step below.

Let $A = (1 - \Delta)^{1/2}$. For a smooth Hilbert-valued function v ,

$$\|v(b)\|_{H^M}^2 - \|v(a)\|_{H^M}^2 = 2 \operatorname{Re} \int_a^b \langle A^{M-1} v_t(t), A^{M+1} v(t) \rangle_{L^2} dt. \quad (3.3.1)$$

Cauchy–Schwarz now places the middle trace between two adjacent controls:

$$v \in L^2(0, T; H^{M+1}), \quad v_t \in L^2(0, T; H^{M-1}), \quad (3.3.2)$$

and the conclusion is

$$v \in C([0, T]; H^M); \quad (3.3.3)$$

the approximation argument and strong representative are proved in Theorem C.3.1.

At $M = 2$, the first concrete pair reads

$$u \in L^2(0, T; H^3), \quad u_t \in L^2(0, T; H^1) \implies u \in C([0, T]; H^2). \quad (3.3.4)$$

The velocity lies above the desired trace, its next temporal response lies below it, and the integral in (3.3.1) brings the two readings to the same time slice.

Apply the same trace to V_n , whose derivative is V_{n+1} . To carry all responses through a finite temporal order N to the common height M , collect exactly the rows used by those pairings:

$$\begin{aligned} \mathfrak{R}_{N,M}(u; T) := & \sum_{n=0}^N \|V_n\|_{L^2(0,T;H^{M+1})}^2 \\ & + \sum_{n=0}^N \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2. \end{aligned} \quad (3.3.5)$$

When $M = 0$, the lower edge is read in H^{-1} . The temporal depth N and spatial height M remain independent. The number $\mathfrak{R}_{N,M}(u; T)$ measures the size of this finite view of the history.

The first nontrivial example, $(N, M) = (1, 2)$, contains two adjacent traces that share V_1 :

Table 3.3.1: The $(1, 2)$ rectangle. The shared response closes the first trace and begins the second.

response	upper row	lower row
$V_0 = u$	$V_0 \in L^2 H^3$	$V_1 \in L^2 H^1$
$V_1 = u_t$	$V_1 \in L^2 H^3$	$V_2 \in L^2 H^1$

The scalar norm does not remember which field produced its entries. Write $\mathcal{R}_{N,M}[u_0; T]$ for this velocity block together with the pressure rows reconstructed by (2.2.6). Every entry is a derivative of the same velocity or of the normalized pressure forced by that velocity. The rectangle is a finite view of one fluid history, not a second object laid over it.

Classical smoothness on a strict subinterval makes each chosen rectangle finite:

$$\mathfrak{R}_{N,M}(u; T) < \infty \quad (T < T_*, N, M < \infty). \quad (3.3.6)$$

The terminal face changes the order of the quantifiers:

$$\forall T < T_* : \quad \mathfrak{R}_{N,M}(u; T) < \infty \quad \not\Rightarrow \quad \sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) < \infty.$$

Fixing (N, M) first leaves one demand: the same bound must survive every cutoff approaching T_* . At the lowest rectangle, that demand can be reduced to one exact critical integral.

3.4 The first rectangle and the exact unpaid integral

The trace argument has reduced an infinite smoothness question to a finite place where the equation can be tested. At $(N, M) = (0, 1)$, the velocity must sit one spatial step above H^1 , and its first time response must sit one step below:

$$u \in L^2(I_*; H^2), \quad u_t \in L^2(I_*; L^2). \quad (3.4.1)$$

Together these two rows reach $u(T_*)$ in H^1 . We now ask the original equation what quantity has to remain finite for this lowest rectangle to reach the terminal face.

Write

$$\begin{aligned} Z(t) &:= \|\Lambda u(t)\|_2^2 = \|\omega(t)\|_2^2, \quad Y(t) := \|\Lambda^2 u(t)\|_2^2 = \|\nabla \omega(t)\|_2^2, \\ D(t) &:= \|\Lambda^{3/2} u(t)\|_2^2. \end{aligned} \tag{3.4.2}$$

The enstrophy balance already contains the spatial row Y ; the only term capable of preventing its integration is vortex stretching. Since the strain is recovered from vorticity by Riesz transforms, Hölder and the Sobolev embeddings give

$$\begin{aligned} \left| \int_{\mathbb{R}^3} \omega \cdot S \omega \, dx \right| &\leq \|\omega\|_2 \|\omega\|_3 \|S\|_6 \\ &\leq C Z^{1/2} D^{1/2} Y^{1/2}. \end{aligned} \tag{3.4.3}$$

Young's inequality returns half of Y to viscosity and leaves the critical coefficient beside Z :

$$Z'(t) + \nu Y(t) \leq C \nu^{-1} D(t) Z(t). \tag{3.4.4}$$

The coefficient has isolated the endpoint interface:

$$D_* := \int_0^{T_*} D(t) \, dt = \int_0^{T_*} \|\Lambda^{3/2} u(t)\|_2^2 \, dt. \tag{3.4.5}$$

Suppose first that $D_* < \infty$. Gronwall applied to (3.4.4) gives

$$\sup_{0 < t < T_*} Z(t) + \int_0^{T_*} Y(t) \, dt < \infty. \tag{3.4.6}$$

Together with the kinetic-energy row, this places $u \in L^2(I_*; H^2)$. The projected momentum equation reads

$$u_t = \nu \Delta u - \mathbb{P}((u \cdot \nabla)u),$$

and its nonlinear part satisfies

$$\|\mathbb{P}((u \cdot \nabla)u)\|_2^2 \leq C \|u\|_6^2 \|\nabla u\|_3^2 \leq C Z D. \tag{3.4.7}$$

The uniform Z roof and the integrable D coefficient place $u_t \in L^2(I_*; L^2)$. This is exactly (3.4.1).

Conversely, a cutoff-uniform $(0, 1)$ rectangle places u in $L^2(I_*; H^2)$ by monotone convergence, and the intermediate $H^{3/2}$ row is then integrable. The lower Sobolev pieces are already paid by the kinetic-energy identity in Proposition A.2.1. We have proved the exact equivalence

$$D_* < \infty \iff \sup_{0 < T < T_*} \mathfrak{R}_{0,1}(u; T) < \infty. \tag{3.4.8}$$

The same interface can be read directly in the critical-height balance. Because $D = 2E_{3/2}$, its time integral is

$$H(t) + \nu \int_0^t D(s) \, ds = H(0) + \int_0^t \mathcal{P}_{1/2}(s) \, ds. \tag{3.4.9}$$

An upper bound on the signed production prefix makes D_* finite because $H \geq 0$. In the other direction, $D_* < \infty$ gives the first rectangle, the adjacent trace in Theorem C.3.1 gives a bounded H , and the same equality bounds the prefix. Thus three descriptions meet at one interface:

$$\sup_{0 < t < T_*} \int_0^t \mathcal{P}_{1/2}(s) ds < \infty \iff D_* < \infty \iff \sup_{0 < T < T_*} \mathfrak{R}_{0,1}(u; T) < \infty. \quad (3.4.10)$$

The signed prefix is not a cheaper route around the first rectangle. It is the same missing payment written in the coordinates of the vortex.

There is a regime in which the datum makes that payment. Set

$$a(t) := \left\| \Lambda^{1/2} u(t) \right\|_2, \quad d(t) := \left\| \Lambda^{3/2} u(t) \right\|_2^2.$$

For $\mathcal{B}(u, u) := \mathbb{P}((u \cdot \nabla)u)$, the dual Sobolev embeddings give

$$\begin{aligned} \|\mathcal{B}(u, u)\|_{\dot{H}^{-1/2}} &\leq C \|u \cdot \nabla u\|_{3/2} \\ &\leq C \|u\|_3 \|\nabla u\|_3 \leq Ca(t)d(t)^{1/2}. \end{aligned} \quad (3.4.11)$$

Pairing this source with the remaining half derivative of the velocity gives

$$|\mathcal{P}_{1/2}(t)| \leq Ca(t)d(t). \quad (3.4.12)$$

The critical-height row becomes

$$\frac{1}{2} \frac{d}{dt} a(t)^2 + (\nu - Ca(t))d(t) \leq 0. \quad (3.4.13)$$

If $a(0) < \nu/C$, continuity prevents a first crossing of the viscous threshold. Before such a crossing, the coefficient is positive and a is nonincreasing, so the crossing cannot occur. Hence

$$\frac{1}{2} a(t)^2 + (\nu - Ca(0)) \int_0^t d(s) ds \leq \frac{1}{2} a(0)^2. \quad (3.4.14)$$

Small critical data pay D_* from the beginning. An arbitrary Schwartz datum need not begin below this threshold, and scaling cannot improve the coefficient: nonlinear production and critical dissipation change with the same weight.

Kinetic energy can still carry a long-lived classical history into the small regime. This produces a horizon beyond which a finite maximal time cannot sit.

Corollary 3.4.1 (Small-data capture horizon). *Let C be the constant in (3.4.13). If a nonzero datum $u_0 \in \mathcal{S}_\sigma$ generates a finite maximal time, then*

$$T_* \leq \frac{C^4 \|u_0\|_2^4}{2\nu^5}. \quad (3.4.15)$$

Proof. Fix $0 < T < T_*$. The kinetic-energy identity in Proposition A.2.1 gives

$$\int_0^T Z(t) dt \leq \frac{\|u_0\|_2^2}{2\nu}.$$

At some $t_0 \in (0, T)$,

$$Z(t_0) \leq \frac{\|u_0\|_2^2}{2\nu T}.$$

Interpolation between the energy and enstrophy heights gives

$$a(t_0)^2 \leq \|u(t_0)\|_2 \|\Lambda u(t_0)\|_2, \quad a(t_0) \leq \frac{\|u_0\|_2}{(2\nu T)^{1/4}}.$$

If $T > C^4 \|u_0\|_2^4 / (2\nu^5)$, the history at t_0 lies below ν/C . Restart the preceding first-crossing argument there:

$$\frac{1}{2}a(t)^2 + (\nu - Ca(t_0)) \int_{t_0}^t d(s) ds \leq \frac{1}{2}a(t_0)^2, \quad t_0 \leq t < T_*.$$

The integral before t_0 is finite by classical smoothness, and the display pays the rest of D_* . Equation (3.4.6) then gives a uniform H^1 roof, hence a uniform L^3 roof, through the alleged last time. This contradicts the endpoint escape already used in (2.1.10). Thus no $T < T_*$ can exceed the displayed horizon. $\square$

For a general datum, the energy balance stops one row below D_* . It pays only

$$\int_0^{T_*} Z(t) dt < \infty.$$

Interpolating $D \leq Z^{1/2}Y^{1/2}$ inside (3.4.3) yields

$$Z'(t) + \nu Y(t) \leq C\nu^{-3}Z(t)^3. \quad (3.4.16)$$

The time integral of Z does not control either $\int Z^3$ or $\sup Z$. The energy and critical calculations have met at the same boundary without crossing it.

They still constrain the shape of any failure. Interpolation between energy and enstrophy gives

$$H(t)^2 = \frac{1}{4} \|\Lambda^{1/2}u(t)\|_2^4 \leq \frac{1}{4} \|u(t)\|_2^2 Z(t),$$

and a second time integration of the energy row gives

$$\int_0^{T_*} H(t)^2 dt \leq \frac{\|u_0\|_2^4}{8\nu}. \quad (3.4.17)$$

Large critical height may occur, but it cannot occupy much time. At the same instant, interpolation between the energy height and the critical dissipation, together with the energy ceiling in Proposition A.2.1, gives, for a nonzero datum,

$$\|\Lambda^{1/2}u(t)\|_2 \leq \|u(t)\|_2^{2/3} \|\Lambda^{3/2}u(t)\|_2^{1/3} \implies D(t) \geq \frac{8H(t)^3}{\|u_0\|_2^4}. \quad (3.4.18)$$

Let the actual history climb from a critical height $M \geq H(0)$ to $2M$. Let t_1 be its first arrival at $2M$, and let $t_0 < t_1$ be its last visit to M before that arrival. On $I_M = (t_0, t_1)$, the height stays between M and $2M$. The occupancy bound derived from Proposition A.2.1 gives

$$|I_M| \leq \frac{\|u_0\|_2^4}{8\nu M^2}. \quad (3.4.19)$$

Meanwhile $|\mathcal{P}_{1/2}| \leq C\sqrt{2H}D$. Integrating the critical balance over this first-passage interval gives

$$M = \int_{I_M} H'(t) dt \leq 2C\sqrt{M} \int_{I_M} D(t) dt, \quad \int_{I_M} D(t) dt \geq \frac{\sqrt{M}}{2C}. \quad (3.4.20)$$

This is the rigorous Zeno remnant. Higher critical records are crossed in shorter intervals, while every completed doubling carries a larger lower cost in critical dissipation. That lower cost exposes the pressure on D_* ; it does not provide the missing upper budget.

Nor can an unpaid budget remain inside a fixed range of frequencies. For $K \geq 1$, set

$$D_{\leq K}(t) := \int_{|\xi| \leq K} |\xi|^3 |\hat{u}(t, \xi)|^2 d\xi.$$

Since $|\xi|^3 \leq K|\xi|^2$ on this region, the kinetic-energy row pays

$$\int_0^{T_*} D_{\leq K}(t) dt \leq K \int_0^{T_*} Z(t) dt \leq \frac{K \|u_0\|_2^2}{2\nu} < \infty. \quad (3.4.21)$$

If D_* diverges, its unpaid part moves beyond every fixed K . The lowest endpoint obstruction is ultraviolet: it belongs to the same narrowing history with which the vortex argument began.

The equivalence (3.4.8) concerns only the lowest rectangle. It does not promote that rectangle through every temporal and spatial order. To carry the complete pressure-coupled Tower to one common terminal datum, the argument below needs the corresponding cutoff-independent control at each finite coordinate. We state that exact remaining input rather than treating algebraic exposure as its proof.

Assumption 3.4.2 (Whole-terminal rectangle condition). Let $u_0 \in \mathcal{S}_\sigma$, let $\nu > 0$, and let (u, p) be the maximal classical solution generated by (u_0, ν) on $[0, T_*)$. If $T_* < \infty$, then for every finite $N, M \in \mathbb{N}_0$,

$$\sup_{0 < T < T_*} \mathfrak{R}_{N,M}(u; T) < \infty. \quad (3.4.22)$$

The pressure rows are generated by (2.2.6) with the fixed decaying normalization.

Here (N, M) is fixed before the supremum over $T < T_*$ is taken. The bound may depend on that finite pair. Every rectangle still belongs to the same history, and a larger rectangle must return the identical derivatives on an overlap.

Assume now that the maximal history has a finite endpoint and that Assumption 3.4.2 holds on this branch. Fix one finite pair (N, M) . Every term in (3.3.5) is the integral of a

nonnegative squared norm, and the intervals $(0, T)$ exhaust I_* . Thus, for any one integrand f ,

$$\int_{I_*} f(t) dt = \lim_{T \uparrow T_*} \int_0^T f(t) dt.$$

The cutoff-independent bound keeps this limit finite for every adjacent row in the chosen rectangle:

$$V_n \in L^2(I_*; H^{M+1}), \quad V_{n+1} = \partial_t V_n \in L^2(I_*; H^{M-1}), \quad 0 \leq n \leq N. \quad (3.4.23)$$

The pressure reconstruction in Proposition C.1.2 uses the Riesz bounds and finite-order Sobolev products to recover the matching rows

$$Q_n \in L^1(I_*; H^M), \quad 0 \leq n \leq N. \quad (3.4.24)$$

The velocity and normalized pressure have now reached the whole interval as one finite pressure-complete rectangle $\mathcal{R}_{N,M}[u_0]$.

Enlarge the rectangle. The fluid does not acquire a second velocity, and a derivative already reached in the smaller rectangle does not change its value. If $N' \leq N$ and $M' \leq M$, let $\rho_{N',M'}^{N,M}$ forget the extra rows and read the retained rows at the lower Sobolev height. Every retained entry is still the same derivative of the same u , so

$$\rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0] = \mathcal{R}_{N',M'}[u_0]. \quad (3.4.25)$$

This equality is the compatibility. It lets every finite view be joined to every larger view without gluing different histories:

$$\mathcal{I}[u_0] := \left(\mathcal{R}_{N,M}[u_0] \right)_{N,M \in \mathbb{N}_0}, \quad \rho_{N',M'}^{N,M} \mathcal{R}_{N,M}[u_0] = \mathcal{R}_{N',M'}[u_0]. \quad (3.4.26)$$

The notation creates no new field. Given a finite temporal order and spatial height, choose one rectangle large enough to see that derivative. Every larger rectangle returns the same derivative of the same u .

Proposition 3.4.3 (The mixed intersection). *Under Assumption 3.4.2, the velocity component of $\mathcal{I}[u_0]$ satisfies*

$$u \in \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I_*; H^m(\mathbb{R}^3)). \quad (3.4.27)$$

In particular,

$$u \in \bigcap_{m=0}^{\infty} L^2(I_*; H^m(\mathbb{R}^3)). \quad (3.4.28)$$

Proof. Fix finite r, m , and choose $N \geq r$ and $M \geq m$. The upper edge of this one rectangle gives

$$\partial_t^n u \in L^2(I_*; H^{M+1}) \hookrightarrow L^2(I_*; H^m) \quad (0 \leq n \leq r).$$

These are the distributional time derivatives of the same velocity field, so $u \in H^r(I_*; H^m)$. Since r and m were arbitrary, every finite coordinate belongs to the intersection; $r = 0$ gives (3.4.28). $\square$

4 One Terminal Datum and the Same-History Restart

The argument has arrived at the face where a smooth history could disappear. On every strict subinterval the same fluid carries every finite derivative. That interior smoothness is not the issue. The issue is whether a fixed finite rectangle keeps one bound while its cutoff moves all the way to T_* . If the bound rises without limit, no terminal datum has been reached. If the whole-terminal condition in Assumption 3.4.2 holds, the adjacent rows reach the face together, their overlaps identify one endpoint, and the local equation can be restarted from the state at which the old history arrived.

This section follows that implication to its end. It does not turn the condition into an arbitrary-data estimate: the unresolved direction remains $(u_0, \nu) \rightarrow D_* < \infty$, already isolated in (3.4.8).

4.1 The intersection reaches one terminal datum

Theorem 4.1.1 (Conditional whole-space continuation). *Let $\nu > 0$, let $u_0 \in \mathcal{S}_\sigma$, and let (u, p) be the maximal classical history of (1.3). Assume its finite branch, if one exists, satisfies Assumption 3.4.2. Then*

$$T_* = \infty, \quad u, p \in C^\infty(\mathbb{R}^3 \times [0, \infty)). \quad (4.1.1)$$

For every $t \geq 0$,

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 ds = \frac{1}{2} \|u_0\|_2^2. \quad (4.1.2)$$

Proof. Suppose instead that the maximal history ends at some finite T_* . The assumed whole-terminal condition carries every finite pressure-complete rectangle across I_* . Whenever two rectangles overlap, (3.4.25) says that they contain the same derivatives of the same history. The mixed intersection in Proposition 3.4.3 gives

$$u \in \bigcap_{r, m < \infty} H^r(I_*; H^m(\mathbb{R}^3)).$$

Hold one spatial height m fixed. The temporal row $r = 1$ places u in $H^1(I_*; H^m)$, so the actual history has an H^m limit as $t \uparrow T_*$. A higher row cannot arrive at a different field: after it is embedded into H^m , both traces are limits of the same $u(t)$. Every finite height reaches one divergence-free endpoint

$$u_* := \lim_{t \uparrow T_*} u(t) \in H_\sigma^\infty(\mathbb{R}^3) \subset \bigcap_{m < \infty} H^m(\mathbb{R}^3). \quad (4.1.3)$$

This is the endpoint furnished by Lemma C.4.1.

The endpoint must remember how the history arrives, not only the velocity at which it arrives. Given finite N and m , the row $r = N + 1$ places $V_N = \partial_t^N u$ in $H^1(I_*; H^m)$. It

has a trace V_N^* , and spatial compatibility puts that trace in H^∞ . The normalized pressure recurrence and the velocity recurrence survive the limit:

$$Q_N^* = R_i R_j \sum_{a=0}^N \binom{N}{a} V_{a,i}^* V_{N-a,j}^*,$$

$$V_{N+1}^* = \nu \Delta V_N^* - \sum_{a=0}^N \binom{N}{a} (V_a^* \cdot \nabla) V_{N-a}^* - \nabla Q_N^*.$$

Thus the entire left-hand velocity jet and the Riesz-normalized pressure jet are generated by u_* , as Proposition C.4.2 records.

Release u_* as initial data at time T_* . The local construction in Lemma C.5.1 gives a right-hand history on $[T_*, T_* + \delta]$. It begins from the same velocity and the same pressure normalization, so its jet agrees with every left-hand trace. The two sides are one smooth solution. Strong uniqueness identifies the right-hand piece with the continuation of the history generated by u_0 , contradicting the maximality of T_* .

The assumed finite branch is impossible, so $T_* = \infty$. The normalized pair is smooth on every finite time interval. The kinetic-energy identity in Proposition A.2.1 now holds on every initial segment of this global history and gives (4.1.2). In particular,

$$\sup_{t \geq 0} \int_{\mathbb{R}^3} |u(x, t)|^2 dx \leq \int_{\mathbb{R}^3} |u_0(x)|^2 dx < \infty.$$

The rapid decay of u_0 pays the last finiteness. Under the whole-terminal condition, this is the global history requested in (1.4). $\square$

The full intersection carries the complete jet to the restart. Its lowest rectangle already contains the shorter endpoint contradiction that explains why the first missing payment is decisive.

4.2 What the first rectangle already knows

Return to the first rectangle, $(N, M) = (0, 1)$. Under the same terminal condition, its adjacent rows are

$$u \in L^2(I_*; H^2), \quad u_t \in L^2(I_*; L^2).$$

Their trace reaches the space between them:

$$u \in C([0, T_*]; H^1).$$

Since $H^1(\mathbb{R}^3) \hookrightarrow L^3(\mathbb{R}^3)$, this gives

$$\sup_{0 \leq t \leq T_*} \|u(t)\|_3 < \infty. \tag{4.2.1}$$

The endpoint theorem of Escauriaza, Seregin, and Šverák says that a finite maximal time forces the opposite behaviour (Escauriaza, G. A. Seregin, and Šverák 2003):

$$T_* < \infty \implies \sup_{0 < t < T_*} \|u(t)\|_3 = \infty. \tag{4.2.2}$$

The rectangle and the endpoint theorem concern the same velocity, the same norm, and the same terminal interval. Their conclusions cannot coexist. The lowest rectangle is already enough to rule out the alleged last time; the higher rectangles do different work by carrying the full jet into the smooth restart.

We can now return to the physical history without beginning another proof. The first rectangle has reached the critical velocity norm. The next rectangle reaches the motion of the critical height and the gradient that stretches the vortex. These consequences tell us what the terminal condition has done to the narrowing mechanism with which the paper began.

4.3 The critical-height roof

Once the H^1 trace reaches the terminal face, the critical height cannot keep climbing underneath it. Indeed, $H = \frac{1}{2} \|\Lambda^{1/2} u\|_2^2$, so

$$H_{\max} := \sup_{0 \leq t \leq T_*} H(t) \leq \frac{1}{2} \sup_{0 \leq t \leq T_*} \|u(t)\|_{H^1}^2 < \infty. \quad (4.3.1)$$

Start at any positive height H_0 , double it successively by $L_k := 2^k H_0$, and count the levels the same history actually reaches:

$$N_{\text{levels}} := \#\{k \in \mathbb{N}_0 : \text{there is a } t \in [0, T_*] \text{ with } H(t) = L_k\}.$$

The roof leaves room for only finitely many such doublings:

$$N_{\text{levels}} \leq \begin{cases} 0, & H_{\max} < H_0, \\ 1 + \lfloor \log_2(H_{\max}/H_0) \rfloor, & H_{\max} \geq H_0. \end{cases} \quad (4.3.2)$$

This conclusion never asks how quickly one level is crossed. The summable heat clocks from Section 2 cannot assemble an infinite ladder because the height itself runs out of levels. A width might still shrink while the height stays bounded, so the geometric picture has one more question to answer.

The next rectangle, $\mathcal{R}_{0,2}$, follows the motion beneath this roof rather than recording only its maximum:

$$u \in L^2(I_*; H^3), \quad u_t \in L^2(I_*; H^1), \quad u \in C([0, T_*]; H^2). \quad (4.3.3)$$

The two adjacent rows now contain both factors in the derivative

$$H'(t) = \operatorname{Re} \left\langle \Lambda^{1/2} u(t), \Lambda^{1/2} u_t(t) \right\rangle_{L^2},$$

and Cauchy–Schwarz pays its total variation:

$$\int_0^{T_*} |H'(t)| dt \leq \|\Lambda^{1/2} u\|_{L_t^2 L_x^2} \|\Lambda^{1/2} u_t\|_{L_t^2 L_x^2} < \infty. \quad (4.3.4)$$

The same rectangle has $E_{3/2} \in L^1(I_*)$. The completed critical row

$$\mathcal{P}_{1/2} = H' + 2\nu E_{3/2}$$

now places the signed production itself in $L^1(I_*)$, and integration records its accumulated action:

$$\int_0^t \mathcal{P}_{1/2}(s) ds = H(t) - H(0) + 2\nu \int_0^t E_{3/2}(s) ds. \quad (4.3.5)$$

The two terms on the right are finite, so

$$A_*(T_*) := \sup_{0 < t < T_*} \int_0^t \mathcal{P}_{1/2}(s) ds < \infty. \quad (4.3.6)$$

The upper prefix in (4.3.6) is already equivalent to the lowest rectangle by (3.4.10); it is the same interface, not a free upstream estimate. The stronger $\mathcal{R}_{0,2}$ control adds finite total variation of H and absolute time integrability of the completed critical row. Sharp rises and falls remain possible, but neither their total height variation nor their signed accumulation can diverge before T_* .

For the vortex, the decisive entry in $\mathcal{R}_{0,2}$ is $u \in L^2(I_*; H^3)$. In three dimensions $H^3 \hookrightarrow W^{1,\infty}$, so

$$\int_0^{T_*} \|\nabla u(t)\|_\infty dt \leq T_*^{1/2} \|u\|_{L^2(I_*; H^3)} < \infty.$$

Both vorticity and strain are parts of this gradient. Their accumulated action before T_* is finite in the precise sense

$$\int_0^{T_*} (\|\nabla u(t)\|_\infty + \|\omega(t)\|_\infty + \|S(t)\|_\infty) dt < \infty. \quad (4.3.7)$$

4.4 The two vortex readings

A drawn narrowing core can refer to two different motions. It may follow the same material particles as they approach one another, or it may keep its visible shape while different particles enter and leave. These pictures are not interchangeable, and neither exhausts possible singular geometry. The terminal rectangles test a precise consequence of each.

Follow the fixed labels first. Let $X(a, t)$ solve

$$\partial_t X(a, t) = u(X(a, t), t), \quad X(a, 0) = a.$$

Two such trajectories satisfy

$$\frac{d}{dt} |X(a, t) - X(b, t)| \geq -\|\nabla u(t)\|_\infty |X(a, t) - X(b, t)|,$$

and the finite Lipschitz action integrates this differential inequality to

$$|X(a, t) - X(b, t)| \geq |a - b| \exp\left(-\int_0^t \|\nabla u(s)\|_\infty ds\right) > 0 \quad (4.4.1)$$

Because the same row also puts u in $L^1(I_*; L^\infty)$, every trajectory has a limit at T_* . Letting $t \uparrow T_*$ preserves the lower bound:

$$|X(a, T_*) - X(b, T_*)| \geq |a - b| \exp\left(-\int_0^{T_*} \|\nabla u(s)\|_\infty ds\right) > 0.$$

Distinct labels remain spatially distinct at the terminal time. If a drawn core collapses to one point, it cannot still contain exactly the material core with which it began.

Even perfect centring cannot rescue a nonzero circulation from this fixed-label picture. The adjacent rows in $\mathcal{R}_{0,3}$, followed by Theorem C.3.1, give

$$u \in C([0, T_*]; H^3).$$

Together with the trace in Theorem C.3.1, the embedding $H^3(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3)$ gives the uniform ceiling

$$L_* := \sup_{0 \leq t \leq T_*} \|\nabla u(t)\|_\infty < \infty. \quad (4.4.2)$$

Relative velocity at any chosen centre $x_c(t)$ obeys

$$|u(t, x) - u(t, x_c)| \leq L_* |x - x_c|.$$

Around a circular cross-section $C_r(t)$, the constant centre velocity integrates to zero. What remains is bounded by its Lipschitz variation:

$$\left| \oint_{C_r(t)} u(t, x) \cdot d\ell \right| \leq 2\pi L_* r^2 \longrightarrow 0 \quad (r \downarrow 0). \quad (4.4.3)$$

As r shrinks, the tangential velocity relative to the centre is $O(r)$ and the circuit has length $O(r)$. Their product is $O(r^2)$, so this centred material core cannot carry a nonzero limiting circulation.

Now let membership change. Particle separation no longer tracks the visible core. A stronger recruitment picture would keep a fixed amount of critical content beyond successively larger frequencies. That precise claim can be tested in the high-frequency tail of the whole velocity field. For $s > 1/2$ and $K \geq 1$, define

$$H_{>K}(t) := \frac{1}{2} \int_{|\xi| > K} |\xi| |\widehat{u}(t, \xi)|^2 d\xi.$$

For every $s > 1/2$, the tail satisfies

$$H_{>K}(t) \leq \frac{1}{2} K^{1-2s} \|u(t)\|_{\dot{H}^s}^2. \quad (4.4.4)$$

Choose one integer $m \geq s$. The compatible assembly and adjacent trace proved in Proposition 3.4.3 and theorem C.3.1, followed by $H^m \hookrightarrow \dot{H}^s$, make the estimate uniform in time and force

$$\lim_{K \rightarrow \infty} \sup_{0 \leq t \leq T_*} H_{>K}(t) = 0. \quad (4.4.5)$$

Recruitment may move activity between scales, but it cannot leave a fixed amount of critical height beyond every cutoff. This excludes the stated persistent tail model. It does not prove that every geometrically narrowing core must produce such a Fourier tail; a separate localization argument would be needed to connect an arbitrary moving core to that frequency statement. What is proved is narrower and exact: fixed labels cannot collapse, their centred circulation vanishes, and the compatible intersection cannot sustain a cutoff-independent critical tail.

4.5 The terminal face carries no local singular content

The first rectangle has already met the global signature of a singular time: the endpoint theorem forces the critical L^3 norm to escape (Escauriaza, G. A. Seregin, and Šverák 2003), while (4.2.1) gives the same velocity a finite roof. The complete intersection lets the same terminal face be read locally.

The mixed velocity traces and the normalized pressure jet in Proposition C.4.2 give the terminal-uniform ceiling

$$K_* := \sup_{0 \leq t \leq T_*} (\|u(t)\|_{L^\infty} + \|p(t)\|_{L^\infty}) < \infty. \quad (4.5.1)$$

Fix a candidate terminal point $z_* = (x_*, T_*)$ and place backward parabolic cylinders at that point:

$$Q_\rho(z_*) = B_\rho(x_*) \times (T_* - \rho^2, T_*).$$

Its spatial volume and time length contribute ρ^5 . Dividing by the critical factor ρ^2 still leaves three powers of the radius:

$$\rho^{-2} \iint_{Q_\rho(z_*)} (|u|^3 + |p|^{3/2}) \, dx \, dt \leq C \rho^3 (K_*^3 + K_*^{3/2}) \longrightarrow 0 \quad (\rho \downarrow 0). \quad (4.5.2)$$

A Caffarelli–Kohn–Nirenberg singular point would retain a positive scale-invariant amount of velocity-pressure content along a shrinking sequence (Caffarelli, Kohn, and Nirenberg 1982). Here the terminal-uniform ceiling makes that content vanish.

We have returned to the same vortex under the whole-terminal condition. Its critical height cannot complete infinitely many doublings; its stretching action is integrable; a fixed material core cannot collapse its labels; the stated persistent-tail recruitment model cannot survive; and every terminal cylinder loses the velocity-pressure content a singular point would require. These are consequences of the smooth terminal datum already used in the same-history restart, not additional routes to an unconditional theorem.

5 Further Consequences and the Euler Boundary

The same finite rows have two further consequences worth keeping. One adds the vorticity across all dyadic scales. The other compares the Navier–Stokes Tower with Euler and isolates the spatial payment that disappears when viscosity is removed.

5.1 A finite sum across infinitely many scales

Fix $T < T_*$. On $[0, T]$, strict-interval parabolic regularity puts u in $L_t^2 H_x^3$; under the whole-terminal condition and the restart above, the same statement is available for every finite T . This one spatial row is enough to add the vorticity from every dyadic wavelength rather than inspect the shells one at a time.

Let $(\Delta_j)_{j \geq -1}$ be an inhomogeneous Littlewood–Paley decomposition (Bahouri, Chemin, and Danchin 2011, Chapter 2). At high frequency, $\Delta_j \omega$ sees wavelengths of order 2^{-j} , while its L^∞ norm records the largest rotation in that shell. Define

$$\|\omega\|_{B_{\infty,1}^0} := \sum_{j \geq -1} \|\Delta_j \omega\|_\infty.$$

Bernstein’s inequality costs three halves of a derivative in passing from L^2 to L^∞ :

$$\|\Delta_j \omega\|_\infty \leq C 2^{3j/2} \|\Delta_j \omega\|_2.$$

The H^2 weight of ω supplies two derivatives. Splitting $2^{3j/2} = 2^{-j/2} 2^{2j}$ exposes the spare half derivative, and Cauchy–Schwarz sums it:

$$\begin{aligned} \sum_{j \geq 0} \|\Delta_j \omega\|_\infty &\leq C \left(\sum_{j \geq 0} 2^{-j} \right)^{1/2} \left(\sum_{j \geq 0} 2^{4j} \|\Delta_j \omega\|_2^2 \right)^{1/2} \\ &\leq C \|\omega\|_{H^2} \leq C \|u\|_{H^3}. \end{aligned}$$

The low shell is controlled directly by $\|\omega\|_2$. Thus one finite Sobolev coordinate pays the full ℓ^1 shell sum:

$$\|\omega\|_{B_{\infty,1}^0} \leq C \|u\|_{H^3}. \quad (5.1.1)$$

Cauchy–Schwarz in time now turns the $L_t^2 H_x^3$ row into

$$\sum_{j \geq -1} \int_0^T \|\Delta_j \omega(t)\|_\infty dt = \int_0^T \|\omega(t)\|_{B_{\infty,1}^0} dt < \infty. \quad (5.1.2)$$

Infinitely many shells may still become active. What cannot happen on the fixed classical interval is a nonsummable succession of their time-integrated peaks. The conclusion uses one finite Sobolev row; it asserts neither an analytic radius nor a weighted estimate over all derivative orders, and its extension to every finite time follows only after the conditional restart has made the history global.

5.2 Why the Euler Tower is different

Remove viscosity while keeping the two histories separate. Both equations can still be differentiated in time, so both possess a mixed jet. Write

$$V_n^{\text{NS}} := \partial_t^n u^{\text{NS}}, \quad Q_n^{\text{NS}} := \partial_t^n p^{\text{NS}},$$

and define $V_n^{\text{E}}, Q_n^{\text{E}}$ from an Euler solution with its own history. The two recurrences differ in one place:

$$V_{n+1}^{\text{NS}} + \sum_{a=0}^n \binom{n}{a} (V_a^{\text{NS}} \cdot \nabla) V_{n-a}^{\text{NS}} + \nabla Q_n^{\text{NS}} = \nu \Delta V_n^{\text{NS}}, \quad (5.2.1)$$

$$V_{n+1}^{\text{E}} + \sum_{a=0}^n \binom{n}{a} (V_a^{\text{E}} \cdot \nabla) V_{n-a}^{\text{E}} + \nabla Q_n^{\text{E}} = 0. \quad (5.2.2)$$

Transport and the nonlocal pressure response remain in both equations. Only the Navier–Stokes row contains $\nu \Delta V_n^{\text{NS}}$, which connects a temporal response to two more spatial derivatives of that same response.

The energy rows expose the same difference. Set

$$E_s^{\text{NS}} := \frac{1}{2} \|\Lambda^s u^{\text{NS}}\|_2^2, \quad E_s^{\text{E}} := \frac{1}{2} \|\Lambda^s u^{\text{E}}\|_2^2,$$

with the corresponding nonlinear transfer terms. Then

$$(E_s^{\text{NS}})' = \mathcal{P}_s^{\text{NS}} - 2\nu E_{s+1}^{\text{NS}}, \quad (E_s^{\text{E}})' = \mathcal{P}_s^{\text{E}}. \quad (5.2.3)$$

Repeated time differentiation of the Navier–Stokes row, followed by division by $(-2\nu)^n$, isolates $E_{n+1/2}^{\text{NS}}$ in (3.1.6). The Euler row still mixes time and space nonlinearly, but no signed coercive term singles out the next spatial height.

If an Euler history were independently known to lie in the analogous all-orders intersection, then every fixed Sobolev norm would be controlled and in particular it would satisfy the Beale–Kato–Majda condition $\int_0^T \|\omega^{\text{E}}(t)\|_\infty dt < \infty$ (Beale, Kato, and Majda 1984). The present theorem supplies no such Euler intersection, and none of its constants is asserted to remain bounded as $\nu \downarrow 0$.

Euler keeps the jet, transport, pressure, and their nonlinear coupling. It loses the viscous rung that places an adjacent higher spatial energy in the Navier–Stokes row. At fixed $\nu > 0$, that rung supplies the strict-interval parabolic gain and exposes the terminal rectangles. If the whole-terminal condition holds, those rectangles reach one datum and the same Navier–Stokes history restarts globally with bounded kinetic energy. What this manuscript does not derive for arbitrary large u_0 is the first cutoff-uniform payment $D_* < \infty$; without it, neither the lowest terminal rectangle nor the global conclusion has been proved.

A Local Construction and Classical Estimates

A maximal history is not merely a chain of short solutions; its last time must be tied to a norm that the equation can test. The local construction makes that link, and the classical estimates show why finite kinetic energy alone cannot keep the history smooth.

A.1 The local construction behind the maximal history

One local theorem must do two jobs without changing its clock. It constructs the initial history and later restarts it from the terminal datum. The clock is set by ν and the datum's H^3 size; any higher regularity already carried by that datum survives on the same interval.

For real s , let

$$H_\sigma^s(\mathbb{R}^3) := \{v \in H^s(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0\}.$$

Proposition A.1.1 (Local construction and continuation alternative). *Fix $t_0 \geq 0$, $\nu > 0$, and $a \in H_\sigma^3(\mathbb{R}^3)$. A time $\delta = \delta(\nu, \|a\|_{H^3}) > 0$ exists for which there is a unique normalized-pressure pair (w, q) satisfying*

$$w \in C([t_0, t_0 + \delta]; H_\sigma^3) \cap L^2(t_0, t_0 + \delta; H^4), \quad w_t \in L^2(t_0, t_0 + \delta; H^2).$$

If the datum also lies in H_σ^m for an integer $m > 3$, this same interval carries

$$w \in C([t_0, t_0 + \delta]; H_\sigma^m) \cap L^2(t_0, t_0 + \delta; H^{m+1}), \quad w_t \in L^2(t_0, t_0 + \delta; H^{m-1}).$$

These local pairs agree on overlaps and form one maximal history. If its maximal time is finite, its H^3 norm is unbounded along times approaching that endpoint.

Proof of Proposition A.1.1. Keep t_0, ν, a fixed as above and measure all real Sobolev orders with

$$\|f\|_{H^s}^2 := \int_{\mathbb{R}^3} \langle \xi \rangle^{2s} |\widehat{f}(\xi)|^2 d\xi, \quad \langle \xi \rangle = (1 + |\xi|^2)^{1/2}.$$

Write $\mathbb{P}$ for the Leray projection and R_j for the Riesz transform with symbol $i\xi_j/|\xi|$. Each acts boundedly on every H^s .

The three rows forced by the nonlinearity and the heat equation

The construction must first absorb one derivative of $w \otimes w$. In three dimensions,

$$H^3(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3), \quad \|fg\|_{H^3} \leq C \|f\|_{H^3} \|g\|_{H^3},$$

so an H^3 velocity is Lipschitz and products stay in H^3 . After the divergence and Leray projection this gives

$$\|\mathbb{P}\nabla \cdot (f \otimes g)\|_{H^2} \leq C \|f\|_{H^3} \|g\|_{H^3}. \quad (\text{A.1.1})$$

Leibniz' rule proves the estimate for integer derivatives, using $H^2 \hookrightarrow L^\infty$ and $H^1 \hookrightarrow L^6$; density then removes the smoothness assumption.

Once the force lies in H^2 , the heat equation places one parabolic derivative above the base state and the time derivative one below it. Thus $z \in L_t^2 H_x^4$ and $z_t \in L_t^2 H_x^2$, with H^3 exactly between them. For any such v ,

$$v \in L^2(J; H^4), \quad v_t \in L^2(J; H^2),$$

frequency truncation followed by the fundamental theorem of calculus and Cauchy–Schwarz yields

$$\left| \|v(t)\|_{H^3}^2 - \|v(r)\|_{H^3}^2 \right| \leq 2 \|v_t\|_{L^2(r,t;H^2)} \|v\|_{L^2(r,t;H^4)}. \quad (\text{A.1.2})$$

The right-hand side vanishes with the length of the time interval. Removing the frequency cutoff therefore gives a unique representative in $C(J; H^3)$, so the three rows of the strong class are

$$C_t H_x^3 \cap L_t^2 H_x^4, \quad \partial_t v \in L_t^2 H_x^2.$$

The continuous middle trace is what lets the equation remember its initial value while the heat operator gains one square-integrable derivative.

A linear estimate aligned with those rows

On $J_\delta = [t_0, t_0 + \delta]$, $0 < \delta \leq 1$, consider

$$z_t - \nu \Delta z = F, \quad z(t_0) = a, \quad F \in L^2(J_\delta; H^2).$$

At the H^3 level the force and solution occupy the adjacent spaces H^2 and H^4 . Their Fourier pairing is

$$\langle F, z \rangle_{H^2, H^4; H^3} := \int_{\mathbb{R}^3} \langle \xi \rangle^6 \widehat{F}(\xi) \cdot \overline{\widehat{z}(\xi)} d\xi.$$

The weight factors as $\langle \xi \rangle^6 = \langle \xi \rangle^2 \langle \xi \rangle^4$ and Cauchy–Schwarz therefore gives

$$|\langle F, z \rangle_{H^2, H^4; H^3}| \leq \|F\|_{H^2} \|z\|_{H^4}. \quad (\text{A.1.3})$$

For a smooth solution the corresponding energy row is

$$\frac{1}{2} \frac{d}{dt} \|z\|_{H^3}^2 + \nu \|\nabla z\|_{H^3}^2 = \text{Re} \langle F, z \rangle_{H^2, H^4; H^3}.$$

The inhomogeneous norm separates the dissipative part exactly:

$$\|z\|_{H^4}^2 = \|z\|_{H^3}^2 + \|\nabla z\|_{H^3}^2.$$

Insert (A.1.3), absorb by Young's inequality, and apply Gronwall to obtain

$$\|z\|_{C(J_\delta; H^3)} + \|z\|_{L^2(J_\delta; H^4)} \leq C_\nu \left(\|a\|_{H^3} + \|F\|_{L^2(J_\delta; H^2)} \right).$$

The equation then controls the missing time row as well:

$$\|z\|_{C(J_\delta; H^3)} + \|z\|_{L^2(J_\delta; H^4)} + \|z_t\|_{L^2(J_\delta; H^2)} \leq C_\nu \left(\|a\|_{H^3} + \|F\|_{L^2(J_\delta; H^2)} \right). \quad (\text{A.1.4})$$

No constant here deteriorates as δ shrinks within $(0, 1]$. Frequency-truncate a and F , solve the smooth equations, and apply (A.1.4) to differences. The Cauchy limit solves the untruncated problem, while (A.1.2) recovers its prescribed strong H^3 value at t_0 . Applying the estimate to the difference of two solutions proves uniqueness.

Closing the nonlinear map

Set

$$X_\delta := C(J_\delta; H_\sigma^3) \cap L^2(J_\delta; H^4), \quad \|w\|_{X_\delta} := \|w\|_{CH^3} + \|w\|_{L^2 H^4}.$$

The projected equation defines the map

$$\Phi(w)(t) := e^{\nu(t-t_0)\Delta} a - \int_{t_0}^t e^{\nu(t-s)\Delta} \mathbb{P} \nabla \cdot (w \otimes w)(s) ds. \quad (\text{A.1.5})$$

All abbreviated time norms below refer to J_δ . The product estimate and the factor $\delta^{1/2}$ from the interval give

$$\|\mathbb{P} \nabla \cdot (w \otimes w)\|_{L^2 H^2} \leq C \delta^{1/2} \|w\|_{CH^3}^2, \quad (\text{A.1.6})$$

$$\|\mathbb{P} \nabla \cdot (w \otimes w - v \otimes v)\|_{L^2 H^2} \leq C \delta^{1/2} (\|w\|_{CH^3} + \|v\|_{CH^3}) \|w - v\|_{CH^3}. \quad (\text{A.1.7})$$

The linear estimate turns these force bounds into

$$\|\Phi(w)\|_{X_\delta} \leq C_\nu \|a\|_{H^3} + C_\nu C \delta^{1/2} \|w\|_{X_\delta}^2$$

and

$$\|\Phi(w) - \Phi(v)\|_{X_\delta} \leq C_\nu C \delta^{1/2} (\|w\|_{X_\delta} + \|v\|_{X_\delta}) \|w - v\|_{X_\delta}.$$

When $a \neq 0$, set $R = 2C_\nu \|a\|_{H^3}$ and choose $\delta \leq 1$ with $4C_\nu C \delta^{1/2} R \leq 1$. Then Φ maps the closed radius- R ball into itself and contracts it by at least a factor of two. For $a = 0$, the fixed point is identically zero. Banach's theorem gives $w \in X_\delta$, with a choice satisfying

$$\delta \geq \min \left\{ 1, c_\nu (1 + \|a\|_{H^3})^{-2} \right\}. \quad (\text{A.1.8})$$

The constant $c_\nu > 0$ depends only on the fixed viscosity and the chosen Sobolev normalization.

Uniqueness is not confined to that ball. If $w, v \in X_\delta$ share the datum, apply the linear and difference estimates to $w - v$ on an initial subinterval short enough to absorb the coefficient. The solutions agree there. Their common endpoint starts the same argument on the next subinterval, and a finite partition covers J_δ . The heat semigroup and $\mathbb{P}$ preserve divergence-free fields; the projected equation also supplies

$$w_t \in L^2(J_\delta; H^2). \quad (\text{A.1.9})$$

Restoring the pressure to the same solution

For the constructed velocity, set

$$q := R_i R_j (w_i w_j). \quad (\text{A.1.10})$$

The algebra property and Riesz bounds imply

$$q \in C(J_\delta; H^3).$$

For $F = \nabla \cdot (w \otimes w)$, the Fourier symbols identify $\mathbb{P}F = F + \nabla q$. Hence the projected fixed point is the pressure-complete solution

$$w_t + (w \cdot \nabla)w + \nabla q = \nu \Delta w, \quad \nabla \cdot w = 0, \quad w(t_0) = a.$$

Membership of $q(t)$ in $H^3(\mathbb{R}^3)$ removes the whole-space additive constant. Two normalized pressures for the same velocity differ by an L^2 function with zero gradient and therefore coincide. The uniqueness already proved is consequently uniqueness of the pair (w, q) .

Higher regularity without a shorter clock

Fix an integer $m \geq 3$ and temporarily take a smooth. Differentiate the projected equation by every ∂^α , $|\alpha| \leq m$, pair with the same derivative of w , and sum. Divergence-free transport cancels at the top order:

$$\left((w \cdot \nabla) \partial^\alpha w, \partial^\alpha w \right)_2 = 0.$$

Every remaining Leibniz term differentiates the transporting factor at least once. The embeddings $H^2 \hookrightarrow L^\infty$ and $H^1 \hookrightarrow L^6$ then give

$$\left(\sum_{|\alpha| \leq m} \|[\partial^\alpha, w \cdot \nabla] w\|_2^2 \right)^{1/2} \leq C_m \|\nabla w\|_\infty \|w\|_{H^m}.$$

Thus the differentiated energy obeys

$$\frac{1}{2} \frac{d}{dt} \|w\|_{H^m}^2 + \nu \|\nabla w\|_{H^m}^2 \leq C_m \|\nabla w\|_\infty \|w\|_{H^m}^2. \quad (\text{A.1.11})$$

The already constructed $C_t H_x^3$ solution has integrable Lipschitz norm on this same interval. Gronwall therefore yields

$$\|w(t)\|_{H^m}^2 \leq \|a\|_{H^m}^2 \exp \left(2C_m \int_{t_0}^t \|\nabla w(s)\|_\infty ds \right). \quad (\text{A.1.12})$$

Retaining the dissipative term gives $w \in L^2(J_\delta; H^{m+1})$. At order $m-1$, the nonlinear term lies in $L^2(J_\delta; H^{m-1})$, so the equation also gives

$$w_t \in L^2(J_\delta; H^{m-1}).$$

The adjacent trace at height m completes the three rows:

$$w \in C(J_\delta; H^m) \cap L^2(J_\delta; H^{m+1}), \quad w_t \in L^2(J_\delta; H^{m-1}). \quad (\text{A.1.13})$$

The pressure formula and the H^m algebra property put q in $C(J_\delta; H^m)$ as well.

For general $a \in H_\sigma^m$, approximate it by divergence-free frequency cutoffs a_k . Their H^3 norms share a bound controlled by $\|a\|_{H^3}$; hence (A.1.8) puts every smooth approximation on one common interval. The fixed-point estimate identifies the H^3 limit with w , while the persistence inequalities bound all three higher rows uniformly. Weak compactness then identifies their limits with the same w . The adjacent trace gives continuity in H^m ; the energy inequality at t_0 , together with weak convergence to a , upgrades the initial convergence to strong convergence in H^m .

The interval never depends on m ; it was chosen once from ν and $\|a\|_{H^3}$. If $a \in H_\sigma^\infty$, repeat the argument at each finite height. Uniqueness in H^3 keeps every height on the same pair, and the pressure formula and momentum equation then recover every finite time derivative. Spatial embedding makes (w, q) smooth throughout the local space–time slab.

Joining the local pieces and locating the only obstruction

Begin at t_0 and extend through every interval allowed by the construction. Strong uniqueness makes the velocity and normalized pressure identical on overlaps, producing one pressure-complete history on a maximal interval $[t_0, T_{\max})$.

Suppose $T_{\max} < \infty$ and

$$\sup_{t_0 \leq t < T_{\max}} \|w(t)\|_{H^3} \leq M.$$

If its displayed H^3 bound were finite, the lifespan estimate would supply the same positive amount of future time

$$\delta_M := \min \left\{ 1, c_\nu (1 + M)^{-2} \right\}$$

from every later datum $w(t)$. Choose $t > T_{\max} - \delta_M/2$. Restarting at $w(t)$ then crosses $T_{\max}$, and uniqueness glues that restart to the supposed maximal history. The contradiction proves

$$T_{\max} < \infty \implies \limsup_{t \uparrow T_{\max}} \|w(t)\|_{H^3} = \infty. \quad (\text{A.1.14})$$

Every higher order already present in the datum persists along each piece, because every piece uses this same H^3 -controlled clock. In particular, $a \in \mathcal{S}_\sigma \subset H_\sigma^\infty$ generates the smooth normalized maximal history claimed above. $\square$

A.2 The Classical Estimates Before Closure

The local theorem has launched a smooth fluid. To keep it alive, we must know which part of that smoothness survives for as long as the solution exists. Kinetic energy survives globally, but at a derivative level too low to restart the classical history. The classical theory measures the exact distance between those two facts.

Energy reaches the endpoint before the gradient does

Proposition A.2.1 (Classical kinetic-energy identity). *Let (u, p) be a sufficiently decaying smooth solution of (1.3). At every time in its interval of existence,*

$$\|u(t)\|_{L^2}^2 + 2\nu \int_0^t \|\nabla u(s)\|_{L^2}^2 ds = \|u_0\|_{L^2}^2. \quad (\text{A.2.1})$$

Proof. Pair the momentum equation with u in space. The transport term carries energy through the fluid but cannot change its total amount:

$$\int_{\mathbb{R}^3} (u \cdot \nabla) u \cdot u \, dx = \frac{1}{2} \int_{\mathbb{R}^3} u \cdot \nabla |u|^2 \, dx = 0.$$

Pressure also does no total work, since

$$\int_{\mathbb{R}^3} \nabla p \cdot u \, dx = - \int_{\mathbb{R}^3} p \nabla \cdot u \, dx = 0.$$

The viscous term alone has a sign, and integration by parts gives

$$\nu \int_{\mathbb{R}^3} \Delta u \cdot u \, dx = -\nu \|\nabla u\|_{L^2}^2.$$

Time integration is exactly (A.2.1). □

Thus the velocity has a pointwise-in-time L^2 ceiling, while the gradient is paid only after its square has been accumulated over time.

Leray's construction preserves precisely this information without assuming the solution remains smooth. Regularization and compactness produce a global weak solution with (Leray 1934)

$$u \in L^\infty(0, T; L^2(\mathbb{R}^3)) \cap L^2(0, T; H^1(\mathbb{R}^3))$$

on each finite interval. The weak history is global, but this estimate by itself neither makes arbitrary three-dimensional Leray solutions unique nor returns the missing smoothness.

The spatial half of the energy class improves through

$$H^1(\mathbb{R}^3) \hookrightarrow L^6(\mathbb{R}^3)$$

and therefore

$$u \in L^2(0, T; L^6(\mathbb{R}^3)).$$

The nonlinear term consumes that gain immediately:

$$\|(u \cdot \nabla) u\|_{L^{3/2}} \leq \|u\|_{L^6} \|\nabla u\|_{L^2},$$

so transport lies in $L_t^1 L_x^{3/2}$, enough for the weak equation to make sense. It does not bound $\|\nabla u(t)\|_2$ at each time. Taller peaks can occupy shorter intervals and still leave the squared time integral finite.

In three dimensions, that missing instantaneous gradient is exactly where a vortex can amplify itself. Put

$$\omega := \nabla \times u.$$

For the divergence-free decaying fields at hand,

$$\|\nabla u\|_{L^2}^2 = \|\omega\|_{L^2}^2,$$

so the time-integrated gradient payment is the time-integrated enstrophy. Taking curl reveals its evolution:

$$\partial_t \omega + (u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nu \Delta \omega.$$

The same velocity gradient that the energy estimate controls only in time now stretches and tilts ω . Pairing this equation with ω gives

$$\frac{1}{2} \frac{d}{dt} \|\omega\|_{L^2}^2 + \nu \|\nabla \omega\|_{L^2}^2 = \int_{\mathbb{R}^3} (\omega \cdot \nabla) u \cdot \omega \, dx.$$

Nothing fixes the sign of the stretching integral, and kinetic energy does not force it below dissipation at each instant. The global estimate has therefore reached the whole interval without reaching the derivative height required for a classical restart.

The derivative height that buys another interval

For an integer $m \geq 0$, collect the first m spatial derivative levels in

$$\|u\|_{H^m}^2 := \sum_{|\alpha| \leq m} \|\partial^\alpha u\|_{L^2}^2,$$

and use the Fourier definition at fractional order. In three dimensions, Sobolev embedding gives (Cerdà 2010, Theorem 7.28)

$$H^s(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3) \quad \text{when} \quad s > k + \frac{3}{2}.$$

In particular,

$$s > \frac{5}{2} \quad \implies \quad H^s(\mathbb{R}^3) \hookrightarrow W^{1,\infty}(\mathbb{R}^3).$$

Above 5/2, one Sobolev norm controls the pointwise gradient that multiplies every differentiated energy row.

Suppose, then, that on $[0, T)$

$$\sup_{0 \leq t < T} \|u(t)\|_{H^s} \leq M \quad \text{for some } s > \frac{5}{2}.$$

At any fixed finite height $m \geq s$, differentiate the equation, pair each row with the matching derivative of u , and sum. Incompressibility cancels the top transport term, leaving

$$\frac{1}{2} \frac{d}{dt} \|u\|_{H^m}^2 + \nu \|\nabla u\|_{H^m}^2 \leq C_m \|\nabla u\|_{L^\infty} \|u\|_{H^m}^2.$$

The assumed H^s roof bounds the coefficient, so Gronwall carries this chosen H^m norm to time T . Its constant may depend on m : smoothness requires each finite height, not one uniform constant for infinitely many heights.

Once the velocity derivatives are present, pressure is forced by them. Taking divergence gives

$$-\Delta p = \sum_{i,j=1}^3 \partial_i \partial_j (u_i u_j),$$

with spatial decay fixing its additive constant. The elliptic relation returns the matching spatial derivatives of p ; the momentum equation then returns

$$\partial_t u = \nu \Delta u - (u \cdot \nabla) u - \nabla p$$

Repeating pressure recovery and the momentum equation produces every fixed mixed derivative. One time-uniform Sobolev roof above $5/2$ therefore recovers the full smooth velocity–pressure pair.

That roof also provides time. Local well-posedness in H^s has a lifespan controlled by s , ν , and the initial H^s size (Fujita and Kato 1964; Kato 1984). Restart at a late slice $u(\cdot, t_0)$. If all such slices remain below M , their lifespans share a positive lower bound; choosing t_0 close enough to T crosses it. Uniqueness joins the restart to the original history. Consequently, a finite maximal time can occur only if

$$\limsup_{t \uparrow T_*} \|u(t)\|_{H^s} = \infty \quad \text{for every } s > \frac{5}{2}.$$

The unpaid distance is now explicit: the energy class gives time-integrated H^1 , while continuation needs time-uniform control above $H^{5/2}$.

The scale on which concentration can hide

The Navier–Stokes scaling explains why energy cannot simply be upgraded to the missing roof. From any solution (u, p) , form

$$u_\lambda(x, t) := \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) := \lambda^2 p(\lambda x, \lambda^2 t)$$

which solves the same equation after space and time are contracted together. Any invariant norm sees the rescaled structure at the same size even while its physical width tends to zero.

The Ladyzhenskaya–Prodi–Serrin line balances the strength of that spatial concentration against the time for which it persists (Prodi 1959; Serrin 1962; Serrin 1963; Ladyzhenskaya 1967). A solution continues through T if

$$u \in L^q(0, T; L^p(\mathbb{R}^3)), \quad \frac{2}{q} + \frac{3}{p} \leq 1, \quad p > 3.$$

At equality, spatial intensity and temporal duration change by inverse factors under the Navier–Stokes scaling; the norm is critical.

The limiting spatial member is $L_t^\infty L_x^3$, since

$$\|u_\lambda(t)\|_{L^3} = \|u(\lambda^2 t)\|_{L^3}.$$

Escauriaza, Seregin, and Šverák showed that even this endpoint condition

$$u \in L^\infty(0, T; L^3(\mathbb{R}^3))$$

continues the solution through T (Escauriaza, G. A. Seregin, and Šverák 2003). Their argument rescales a first singularity into an ancient limit and eliminates it by backward uniqueness. Hence a finite last time must make the critical L^3 norm escape.

The same threat has a local description. Around $z_0 = (x_0, t_0)$, use the parabolic cylinder

$$Q_r(z_0) := B_r(x_0) \times (t_0 - r^2, t_0).$$

Caffarelli, Kohn, and Nirenberg proved that small scale-invariant velocity and pressure content on $Q_r(z_0)$ makes a smaller cylinder regular (Caffarelli, Kohn, and Nirenberg 1982). A singular point must therefore retain a definite amount of that content at every sufficiently small scale. Their covering argument makes the possible singular set small in parabolic Hausdorff measure, but one surviving point is enough to destroy global smoothness.

For smooth compactly supported whole-space data, Seregin sharpens the endpoint statement to actual divergence along the terminal approach (G. Seregin 2012):

$$\lim_{t \uparrow T_*} \|u(t)\|_{L^3(\mathbb{R}^3)} = \infty.$$

Tao's quantitative refinement forces a positive power of a triple logarithm of the inverse time remaining along a sequence of terminal times (Tao 2021). This lower rate is far slower than any power law; the classical results still permit much faster concentration.

The one estimate still absent

If the maximal time were finite, kinetic energy would remain bounded while the critical L^3 norm and every continuation-grade Sobolev norm escaped. Locally, some parabolic cylinders would retain nonzero scale-invariant content at arbitrarily small radii. These results describe what a singularity must do; they do not give the terminal-uniform estimate that prevents it.

The mismatch with energy is visible in the two scaling laws

$$\|u_\lambda(t)\|_{L^3} = \|u(\lambda^2 t)\|_{L^3}, \quad \|u_\lambda(t)\|_{L^2}^2 = \lambda^{-1} \|u(\lambda^2 t)\|_{L^2}^2.$$

A shorter structure retains its critical L^3 size while carrying less L^2 energy. A global kinetic-energy ceiling therefore cannot become a scale-independent ceiling on the critical concentration.

The same scaling shortens the clock. Repeatedly halving the active width gives parabolic durations proportional to

$$1, \frac{1}{4}, \frac{1}{16}, \dots,$$

whose sum is finite. Each stage can retain critical size while occupying less time, and none of the estimates above forbids that arithmetic. This is all the Zeno schedule asserts: it exposes an available timing, not a constructed singular Navier–Stokes history.

Closure therefore asks for one terminal-uniform continuation-grade norm on the actual maximal history.

B Derivative and Finite-Record Calculus

Strict classicality gives every derivative before T_* , but it does not tell us which finite part of that information can survive uniformly as the cutoff approaches the endpoint. The calculus here turns the derivative family into finite records so the terminal estimate has a precise object to keep bounded.

B.1 Critical Height at a Finite Endpoint

Return to the one maximal solution generated by the rapidly decaying datum u_0 at fixed $\nu > 0$, and suppose its last time T_* is finite. The field is smooth before that time. Its threat must therefore be a change of scale along the same history, not an already singular initial state.

Kinetic energy remains bounded, yet a structure’s contribution to that energy changes when its width changes. For $\lambda > 0$, set

$$u_\lambda(x, t) := \lambda u(\lambda x, \lambda^2 t), \quad p_\lambda(x, t) := \lambda^2 p(\lambda x, \lambda^2 t)$$

These fields solve Navier–Stokes with the same viscosity on the rescaled time interval. Space contracts by λ^{-1} , time by λ^{-2} , and velocity grows by λ ; one time derivative and the viscous two-space- derivative term consequently transform together.

Let $\Lambda = (-\Delta)^{1/2}$ and measure height by

$$\|v\|_{\dot{H}^s}^2 := \int_{\mathbb{R}^3} |\xi|^{2s} |\widehat{v}(\xi)|^2 d\xi.$$

Proposition B.1.1 (Critical Sobolev scaling). *Whenever the homogeneous norm is finite, every $s \in \mathbb{R}$ and $\lambda > 0$ satisfy*

$$\|u_\lambda(t)\|_{\dot{H}^s}^2 = \lambda^{2s-1} \|u(\lambda^2 t)\|_{\dot{H}^s}^2.$$

The only scale-invariant member is $s = \frac{1}{2}$.

Proof. The rescaled Fourier transform is

$$\widehat{u_\lambda}(\xi, t) = \lambda^{-2} \widehat{u}(\xi/\lambda, \lambda^2 t).$$

Substitution into the norm, followed by $\eta = \xi/\lambda$, gives

$$\begin{aligned}\|u_\lambda(t)\|_{\dot{H}^s}^2 &= \int_{\mathbb{R}^3} |\xi|^{2s} \lambda^{-4} \left| \widehat{u}(\xi/\lambda, \lambda^2 t) \right|^2 d\xi \\ &= \lambda^{2s-1} \int_{\mathbb{R}^3} |\eta|^{2s} \left| \widehat{u}(\eta, \lambda^2 t) \right|^2 d\eta.\end{aligned}$$

The prefactor is one exactly when $2s - 1 = 0$, proving the claim. $\square$

Compression therefore tilts every Sobolev height except $1/2$. Name the quantity that remains unchanged:

$$H(t) := \frac{1}{2} \|u(t)\|_{\dot{H}^{1/2}}^2 = \frac{1}{2} \left\| \Lambda^{1/2} u(t) \right\|_2^2.$$

An exact rescaling can now narrow and accelerate a smooth configuration without changing H .

At energy height, the same narrowing has the opposite weight:

$$\|u_\lambda(t)\|_2^2 = \lambda^{-1} \left\| u(\lambda^2 t) \right\|_2^2.$$

A finer structure can therefore retain its critical height while carrying less kinetic energy. The energy ceiling does not itself prevent the active width from continuing to fall.

The parabolic durations in Section A.2 allow successively finer scales to fit arithmetically before a finite endpoint. At critical height, those rescaled copies can all have the same H while their energy falls. They are still only copies: scaling has not made the actual solution visit them. Such a passage would have to be generated by transport, pressure, incompressibility, and viscosity acting together on the preceding state.

The scale is now fixed. A finite endpoint turns the remaining issue into response: if one row of the actual field escapes, what must the next time row do before the available time runs out?

B.2 What finite time implies after one response escapes

The equation supplies the temporal rows; finite time supplies the implication between two adjacent rows. If one response has no finite roof while the next is integrable, the fundamental theorem of calculus would nevertheless give the first a terminal limit. The following argument records that contradiction. It propagates escape through rows already known to exist and creates none of them.

Use one fixed Banach space B throughout and write the Eulerian rows as

$$V_n(t) := \partial_t^n u(\cdot, t),$$

Assume $V_n : [0, T_*) \rightarrow B$ is locally absolutely continuous and has Bochner derivative V_{n+1} . For $0 \leq s < t < T_*$,

$$V_n(t) - V_n(s) = \int_s^t V_{n+1}(\tau) d\tau$$

and therefore

$$\|V_n(t) - V_n(s)\|_B \leq \int_s^t \|V_{n+1}(\tau)\|_B d\tau.$$

Suppose

$$\sup_{0 < t < T_*} \|V_n(t)\|_B = \infty,$$

Since $\|V_n(0)\|_B$ is finite, the integral on the right cannot remain finite; hence

$$\int_0^{T_*} \|V_{n+1}(\tau)\|_B d\tau = \infty.$$

The interval itself is finite, so an L^1 divergence forces

$$\sup_{0 < t < T_*} \|V_{n+1}(t)\|_B = \infty.$$

Iteration is licensed only when every successive pair has this same fixed space, local absolute continuity, and Bochner-derivative identity. Under those hypotheses, finite time turns escape of one temporal response into escape of the next.

B.3 The Pressure-Complete Mixed Jet

At temporal order n , the equation keeps four pieces together: V_{n+1} , $\nu\Delta V_n$, every temporal split of transport, and the normalized pressure row Q_n . A single spatial derivative already shows how this row widens:

$$\partial_\ell((V_a \cdot \nabla)V_{n-a}) = ((\partial_\ell V_a) \cdot \nabla)V_{n-a} + (V_a \cdot \nabla)\partial_\ell V_{n-a}.$$

At multi-index order α , every derivative must be distributed between the transporting and transported factors. Denote the resulting coordinates by

$$J_{n,\alpha} := \partial_t^n \partial_x^\alpha u, \quad \Pi_{n,\alpha} := \partial_t^n \partial_x^\alpha p,$$

and use

$$\binom{\alpha}{\beta} := \prod_{k=1}^3 \binom{\alpha_k}{\beta_k}.$$

The two Leibniz rules then give the complete transport split

$$\partial_t^n \partial_x^\alpha ((u \cdot \nabla)u) = \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta}.$$

Substitution keeps that split beside viscosity and pressure:

$$\begin{aligned} J_{n+1,\alpha} + \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta} \\ + \nabla \Pi_{n,\alpha} = \nu \Delta J_{n,\alpha}, \quad \nabla \cdot J_{n,\alpha} = 0. \end{aligned}$$

Taking divergence supplies the corresponding pressure row:

$$-\Delta \Pi_{n,\alpha} = \partial_i \partial_j \sum_{a=0}^n \sum_{\beta \leq \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta})_i (J_{n-a,\alpha-\beta})_j.$$

Every displayed coordinate is a derivative of the original pressure-complete pair. A finite rectangle will retain only finitely many of them, but it never changes the underlying solution.

The viscous coupling links the next temporal row to two additional spatial derivatives of the present row. The critical-height energy calculation isolates that ascent while preserving the transport and pressure terms that travel with it.

B.4 The Spatial Column Inside Critical Height

To follow the viscous link from a temporal response into space, take the scalar energy at an arbitrary spatial height $s \geq 0$:

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2,$$

In particular,

$$H(t) = E_{1/2}(t).$$

Keeping s free lets the same calculation be reused after the critical row.

Apply Λ^s to the full momentum equation and pair with $\Lambda^s u$. The temporal contribution is

$$\langle \Lambda^s u, \Lambda^s \partial_t u \rangle = \frac{d}{dt} \frac{1}{2} \|\Lambda^s u\|_2^2 = E'_s.$$

Viscosity commutes with Λ^s , and spatial decay gives

$$\nu \langle \Lambda^s u, \Lambda^s \Delta u \rangle = -\nu \|\Lambda^{s+1} u\|_2^2 = -2\nu E_{s+1}.$$

Pressure cancels in this global scalar balance only after integration by parts uses incompressibility:

$$\langle \Lambda^s u, \Lambda^s \nabla p \rangle = -\langle \nabla \cdot \Lambda^s u, \Lambda^s p \rangle = 0.$$

It remains present in the local equation and in every mixed coordinate. Accordingly, keep the signed transfer in its pressure-complete form:

$$\mathcal{P}_s(t) := -\left\langle \Lambda^s u, \Lambda^s \left((u \cdot \nabla) u + \nabla p \right) \right\rangle.$$

The resulting identity is

$$E'_s = \mathcal{P}_s - 2\nu E_{s+1}.$$

One derivative in time has reached one full Sobolev level higher in space, but only together with the signed transfer at the original height.

At $s = 1/2$, this becomes

$$H' = \mathcal{P}_{1/2} - 2\nu E_{3/2}.$$

The scale-invariant height now touches $E_{3/2}$. Its mere appearance is not a bound: transfer and dissipation can cancel inside H' . Differentiate the whole row, not the isolated dissipative term.

The next energy row is

$$E'_{3/2} = \mathcal{P}_{3/2} - 2\nu E_{5/2},$$

and substitution into the derivative of the critical row yields

$$\begin{aligned} H'' &= \mathcal{P}'_{1/2} - 2\nu E'_{3/2} \\ &= \mathcal{P}'_{1/2} - 2\nu \mathcal{P}_{3/2} + (2\nu)^2 E_{5/2}. \end{aligned}$$

The response H'' reaches $E_{5/2}$, while retaining the transfer from each preceding height. One more derivative uses

$$E'_{5/2} = \mathcal{P}_{5/2} - 2\nu E_{7/2}$$

and gives

$$H''' = \mathcal{P}''_{1/2} - 2\nu \mathcal{P}'_{3/2} + (2\nu)^2 \mathcal{P}_{5/2} - (2\nu)^3 E_{7/2}.$$

The fourth response continues the same recurrence:

$$H^{(4)} = \mathcal{P}'''_{1/2} - 2\nu \mathcal{P}''_{3/2} + (2\nu)^2 \mathcal{P}'_{5/2} - (2\nu)^3 \mathcal{P}_{7/2} + (2\nu)^4 E_{9/2}.$$

At each step, the highest spatial energy rises once and the entire lower transfer history remains in the row.

This produces a triangular identity.

Proposition B.4.1 (Critical-height spatial ascent). *For every integer $n \geq 1$,*

$$H^{(n)} = (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{j+1/2}.$$

Proof. The critical balance is the case $n = 1$. If the formula holds at order n , differentiate it and use

$$E'_{n+1/2} = \mathcal{P}_{n+1/2} - 2\nu E_{n+3/2}.$$

The new transfer extends the sum by one term, while the dissipative part becomes $(-2\nu)^{n+1} E_{n+3/2}$. This is the formula at order $n + 1$. $\square$

The proposition is an identity, not an estimate. Knowing $H^{(n)}$ cannot isolate $E_{n+1/2}$ until the complete signed transfer sum is controlled.

Critical scaling was homogeneous; continuation uses inhomogeneous Sobolev spaces. For $s > 0$, the two are joined by

$$\|u(t)\|_{H^s}^2 \simeq \|u(t)\|_2^2 + \|\Lambda^s u(t)\|_2^2 = \|u(t)\|_2^2 + 2E_s(t).$$

The kinetic-energy identity pays the low-frequency term:

$$\frac{1}{2} \|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(\tau)\|_2^2 d\tau = \frac{1}{2} \|u_0\|_2^2.$$

Hence a bound for $E_s(t)$ completes an H^s bound, while time-integrability of E_s completes the high-frequency part of $L_t^2 H_x^s$ on a finite interval.

The ascent begins at $1/2$ because scaling selected it, and the first response reaches $3/2$. At the terminal face the logic reverses: whole-terminal Sobolev rows first bound these two heights, from which H and its first dissipative level follow.

Each finite number of responses reaches only a finite spatial height. To keep every finite coordinate of the derivative family, the same velocity must occupy every finite temporal and spatial Sobolev row.

B.5 Simultaneous Sobolev Intersections

The ascent has no final spatial height, although each concrete derivative request uses only finitely many rungs. To make all $\partial_x^\alpha u$ with $|\alpha| \leq k$ continuous, use the three-dimensional embedding

$$H^m(\mathbb{R}^3) \hookrightarrow C^k(\mathbb{R}^3)$$

whenever

$$m > k + \frac{3}{2}.$$

Thus each requested order k chooses one finite height above its threshold. There is no largest m as k ranges over all finite orders, but no single infinite-order norm is needed. The proper object is the coordinatewise intersection

$$\bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3),$$

whose coordinates are all measured on the same field. For each fixed k , one coordinate supplies C^k , hence

$$\bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3) \hookrightarrow C^\infty(\mathbb{R}^3).$$

Time behaves the same way. Let $I = (0, T)$, $T < T_*$, and let X be a Hilbert space.

The vector-valued embedding

$$H^r(I; X) \hookrightarrow C^q(\bar{I}; X)$$

holds for

$$r > q + \frac{1}{2}.$$

Each requested time derivative q chooses one finite r , and all finite requests give

$$\bigcap_{r=0}^{\infty} H^r(I; X) \hookrightarrow C^\infty(\bar{I}; X).$$

No common bound across all r is asserted.

The mixed jet makes the temporal and spatial choices simultaneously. For example, request

$$\partial_t^2 \partial_x^\alpha u, \quad |\alpha| \leq 4.$$

The one membership

$$u \in H^3(I; H^6(\mathbb{R}^3))$$

places this entire finite request in

$$C^2(\bar{I}; C^4(\mathbb{R}^3)),$$

since

$$3 > 2 + \frac{1}{2}, \quad 6 > 4 + \frac{3}{2}.$$

The pair $(2, 4)$ is only an example. For arbitrary finite q, k , choose finite indices with

$$r > q + \frac{1}{2}, \quad m > k + \frac{3}{2}.$$

Then

$$H^r(I; H^m(\mathbb{R}^3)) \hookrightarrow C^q(\bar{I}; C^k(\mathbb{R}^3)).$$

Collect every finite time-space request on the same velocity by defining

$$\mathcal{V}_\infty(I) := \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I; H^m(\mathbb{R}^3)).$$

Explicitly,

$$u \in \mathcal{V}_\infty(I) \iff \|u\|_{H^r(I; H^m)} < \infty \text{ for every finite pair } (r, m).$$

The norms remain separate coordinates; the intersection does not sum them or replace u by a family of fields. The finite embeddings imply

$$\mathcal{V}_\infty(I) \hookrightarrow C^\infty(\bar{I} \times \mathbb{R}^3).$$

Choosing one temporal height above the trace threshold and one spatial height above $3/2$ places both critical levels on the closed interval:

$$u(t) \in H^{1/2}(\mathbb{R}^3) \quad \text{and} \quad u(t) \in H^{3/2}(\mathbb{R}^3)$$

Their scalar quantities are consequently defined there by

$$H(t) = \frac{1}{2} \left\| \Lambda^{1/2} u(t) \right\|_2^2, \quad E_{3/2}(t) = \frac{1}{2} \left\| \Lambda^{3/2} u(t) \right\|_2^2$$

Scaling and the first dissipative ascent selected these two heights; the intersection is what carries them to the terminal face.

Pressure is recovered alongside this velocity space, not inserted into it. At finite temporal depth write

$$Q_n = \partial_t^n p,$$

The Poisson equation recovers Q_n from the velocity rows, while the differentiated momentum equation keeps ∇Q_n beside response, viscosity, and transport. This produces a pressure-complete jet without misidentifying pressure as a velocity coordinate.

An actual estimate will control only a finite block. The two adjacent Sobolev levels required for a common time trace determine that block's upper and lower edges.

B.6 Finite Adjacent Rectangles

Only finite estimates can build the intersection. Begin on $I = (0, T)$, $T < T_*$, with the first request that needs adjacent rows: bring both

$$V_0 = u \quad \text{and} \quad V_1 = \partial_t u$$

to one time slice in $H^2(\mathbb{R}^3)$. The upper spatial row $V_0 \in L_t^2 H_x^3$ alone has no time trace. Its lower adjacent row is

$$\partial_t V_0 = V_1 \in L^2(I; H^1).$$

To trace V_1 at that same height, repeat the pair once:

$$V_1 \in L^2(I; H^3), \quad \partial_t V_1 = V_2 \in L^2(I; H^1).$$

Together these are

$$V_0, V_1 \in L^2(I; H^3), \quad V_1, V_2 \in L^2(I; H^1).$$

The repeated V_1 plays two roles: it closes the lower side of the first trace and occupies the upper side of the second.

The reason the levels straddle H^2 is visible directly. Let

$$A := (1 - \Delta)^{1/2}, \quad \|v\|_{H^s} = \|A^s v\|_2.$$

For smooth v ,

$$\|v(t)\|_{H^2}^2 - \|v(s)\|_{H^2}^2 = 2 \operatorname{Re} \int_s^t \left\langle A \partial_\tau v(\tau), A^3 v(\tau) \right\rangle_{L^2} d\tau.$$

The integrand is bounded by

$$\left| \left\langle A \partial_\tau v, A^3 v \right\rangle_{L^2} \right| \leq \|\partial_\tau v\|_{H^1} \|v\|_{H^3},$$

which is integrable in time whenever

$$v \in L^2(I; H^3), \quad \partial_t v \in L^2(I; H^1).$$

Approximation and weak continuity consequently give

$$v \in C([0, T]; H^2(\mathbb{R}^3)).$$

Applying this once to each adjacent pair yields

$$V_0, V_1 \in C([0, T]; H^2(\mathbb{R}^3)).$$

Collect the four norms consumed by these two traces in

$$\begin{aligned} \mathfrak{R}_{1,2}(u; T) &:= \|u\|_{L^2(0,T;H^3)}^2 + \|\partial_t u\|_{L^2(0,T;H^3)}^2 \\ &\quad + \|\partial_t u\|_{L^2(0,T;H^1)}^2 + \|\partial_t^2 u\|_{L^2(0,T;H^1)}^2. \end{aligned}$$

All four entries belong to the original solution. Their grouping is a record of what the two traces consume, not a construction of four new fields.

Now choose independent finite cutoffs $N, M \in \mathbb{N}_0$. To trace each retained temporal row at H^M , use the adjacent scale

$$H^{M+1}(\mathbb{R}^3) \hookrightarrow H^M(\mathbb{R}^3) \hookrightarrow H^{M-1}(\mathbb{R}^3),$$

with H^{-1} as the lower member when $M = 0$. For every $0 \leq n \leq N$, the required pair is

$$V_n \in L^2(0, T; H^{M+1}), \quad V_{n+1} = \partial_t V_n \in L^2(0, T; H^{M-1}).$$

Their squared finite velocity norm is

$$\begin{aligned} \mathfrak{R}_{N,M}(u; T) &:= \sum_{n=0}^N \|V_n\|_{L^2(0,T;H^{M+1})}^2 \\ &\quad + \sum_{n=0}^N \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2. \end{aligned}$$

For a smooth representative, each row satisfies

$$\left| \frac{d}{dt} \|V_n(t)\|_{H^M}^2 \right| \leq 2 \|V_{n+1}(t)\|_{H^{M-1}} \|V_n(t)\|_{H^{M+1}}.$$

Its right-hand side is integrable, so density and weak continuity yield

$$V_n \in C([0, T]; H^M(\mathbb{R}^3)), \quad 0 \leq n \leq N.$$

The temporal cutoff N chooses how many responses are retained; the spatial cutoff M chooses their common trace height. They remain independent, and the rectangle constant may depend on either one.

The scalar $\mathfrak{R}_{N,M}(u; T)$ remembers only the velocity norms. A rectangle must also remember why those rows belong together: retain the normalized pressure coordinates and the differentiated equations. For the pair generated by u_0 , define

$$\mathcal{R}_{N,M}[u_0; T] := \left((V_n)_{n=0}^{N+1}, (Q_n)_{n=0}^N; \mathfrak{R}_{N,M}(u; T) \right),$$

with

$$V_n = \partial_t^n u, \quad Q_n = \partial_t^n p.$$

Each retained temporal row obeys

$$V_{n+1} + \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} + \nabla Q_n = \nu \Delta V_n,$$

$$\nabla \cdot V_n = 0,$$

and

$$-\Delta Q_n = \partial_i \partial_j \sum_{a=0}^n \binom{n}{a} (V_a)_i (V_{n-a})_j.$$

Rapid decay fixes every pressure constant. At the retained depths, the mixed velocity coordinates are the weak derivatives

$$J_{n,\alpha} = \partial_x^\alpha V_n,$$

and the pressure coordinates are the corresponding derivatives of Q_n . Thus $\mathfrak{R}_{N,M}$ measures the finite velocity block, whereas $\mathcal{R}_{N,M}$ retains its pressure-complete equation identity.

To restrict such a block in time, let $J = (0, \tau)$, $0 < \tau \leq T_*$, and let $\mathcal{X}_{N,M}(J)$ consist of records with

$$W_n \in L^2(J; H^{M+1}), \quad W_{n+1} \in L^2(J; H^{M-1}) \quad (0 \leq n \leq N),$$

with $\partial_t W_n = W_{n+1}$ distributionally on $J \times \mathbb{R}^3$, and normalized pressure rows

$$P_n = R_i R_j \sum_{a=0}^n \binom{n}{a} (W_a)_i (W_{n-a})_j \in L^1(J; H^M) \quad (0 \leq n \leq N),$$

Their gradients lie in $L^1(J; H^{M-1})$, and the displayed momentum and divergence equations hold distributionally. The scalar component is the adjacent-row norm of the retained velocities on J . In particular,

$$\mathcal{R}_{N,M}[u_0; T] \in \mathcal{X}_{N,M}((0, T)).$$

As (N, M) ranges over finite pairs, the rectangles cover every finite coordinate of the intersection, each with its own finite bound. Overlapping rectangles contain the same distribution wherever they name the same derivative. That literal agreement is the datum needed for terminal assembly.

C Compatible Terminal Assembly

A bound for every finite record is still not a terminal solution. Because the records are obtained separately, their shared coordinates must remain the same as the rectangle and cutoff change; otherwise their limits need not belong to one history.

C.1 The Terminal Quantifier

Fix again the rapid-decay datum u_0 , the viscosity $\nu > 0$, and their single maximal normalized history (u, p) . Suppose its maximal time is finite and write $I_* = (0, T_*)$. Every coordinate below is a derivative of this history. The datum belongs to the smooth solenoidal class

$$H_\sigma^\infty(\mathbb{R}^3) := \left\{ v \in \bigcap_{m \in \mathbb{N}_0} H^m(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot v = 0 \right\}. \quad (\text{C.1.1})$$

Spatial decay fixes pressure from the velocity without an additive constant. With $R_j = \partial_j(-\Delta)^{-1/2}$,

$$-\Delta p = \partial_i \partial_j (u_i u_j), \quad p = R_i R_j (u_i u_j). \quad (\text{C.1.2})$$

Direct time differentiation, with $V_n = \partial_t^n u$ and $Q_n = \partial_t^n p$, gives

$$Q_n = R_i R_j \sum_{a=0}^n \binom{n}{a} V_{a,i} V_{n-a,j}, \quad (\text{C.1.3})$$

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^n \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n. \quad (\text{C.1.4})$$

Every division of the time derivatives between the two transport factors remains in the binomial sum. Thus each temporal row still contains the full transport, pressure, viscosity, and divergence-free evolution.

Choose finite temporal depth N , spatial height M , and a strict cutoff $T < T_*$. The adjacent-row norm is

$$\begin{aligned} \mathfrak{R}_{N,M}(u; T) := & \sum_{n=0}^N \left\| \partial_t^n u \right\|_{L^2(0,T; H^{M+1})}^2 \\ & + \sum_{n=0}^N \left\| \partial_t^{n+1} u \right\|_{L^2(0,T; H^{M-1})}^2. \end{aligned} \quad (\text{C.1.5})$$

When $M = 0$, interpret the lower row in the inhomogeneous Bessel-potential space H^{-1} . The record $\mathcal{R}_{N,M}[u_0; T]$ retains these velocity norms, their normalized pressure rows, and the equations that identify all entries as coordinates of (u, p) . Smoothness pays every finite record once $T < T_*$ has been fixed.

Lemma C.1.1 (Rectangles on strict classical subintervals). *For each strict cutoff $T < T_*$ and each finite pair N, M ,*

$$\mathfrak{R}_{N,M}(u; T) < \infty. \quad (\text{C.1.6})$$

If $N' \leq N$, $M' \leq M$, and $T' \leq T$, forgetting the excess coordinates and restricting time produces the smaller record.

Proof. The closed interval $[0, T]$ stays strictly inside the classical lifespan. Persistence puts every fixed $\partial_t^n u$ continuously in every fixed H^q there, and compactness bounds that coordinate. Hence

$$\|\partial_t^n u\|_{L^2(0, T; H^q)}^2 \leq T \sup_{0 \leq t \leq T} \|\partial_t^n u(t)\|_{H^q}^2 < \infty.$$

Apply this inequality to the finitely many orders in (C.1.5). Restriction changes neither the solution nor its derivatives, which also proves compatibility. $\square$

Strict-cutoff finiteness is not terminal control. The scalar function $f(t) = (T_* - t)^{-1/2}$ already separates the two quantifiers:

$$f \in L^2(0, T) \quad \text{for every } T < T_*, \quad f \notin L^2(I_*).$$

This is not a fluid model; it only shows that every strict interval may be finite while the terminal integral diverges.

The preceding lemma permits exactly that growth. The whole-terminal condition in Assumption 3.4.2 is the additional input: after a finite (N, M) is chosen, its bound is independent of the cutoff. For $N' \leq N$ and $M' \leq M$, define the terminal bonding map

$$\rho_{N', M'; I_*}^{N, M; I_*} : \mathcal{X}_{N, M}(I_*) \longrightarrow \mathcal{X}_{N', M'}(I_*)$$

by dropping the higher rows, embedding the retained velocity and pressure coordinates into the lower spaces, and recomputing the adjacent-row norm on I_* . Direct restriction and restriction in stages agree, while an unchanged pair gives the identity.

Proposition C.1.2 (Terminal realization of the rectangle estimate). *Assume Assumption 3.4.2. For every finite N, M , the pressure-complete history satisfies*

$$\begin{aligned} \mathfrak{R}_{N,M}(u; I_*) &:= \sum_{n=0}^N \|\partial_t^n u\|_{L^2(I_*; H^{M+1})}^2 \\ &\quad + \sum_{n=0}^N \|\partial_t^{n+1} u\|_{L^2(I_*; H^{M-1})}^2 < \infty. \end{aligned} \tag{C.1.7}$$

The same coordinates therefore define a record on all of I_* :

$$\mathcal{R}_{N,M}[u_0] \in \mathcal{X}_{N,M}(I_*),$$

For every lower pair (N', M') , its terminal restriction satisfies

$$\rho_{N',M';I_*}^{N,M;I_*}(\mathcal{R}_{N,M}[u_0]) = \mathcal{R}_{N',M'}[u_0]. \tag{C.1.8}$$

All shared entries are the same distributional derivatives of u , and all pressure entries retain the normalization (C.1.2).

Proof. Fix the pair (N, M) and, for $0 \leq n \leq N$, set

$$f_n^+(t) := \|V_n(t)\|_{H^{M+1}}^2, \quad f_n^-(t) := \|V_{n+1}(t)\|_{H^{M-1}}^2.$$

The functions are nonnegative and $(0, T)$ increases to I_* . Monotone convergence gives, for either sign,

$$\int_{I_*} f_n^\sigma(t) dt = \lim_{T \uparrow T_*} \int_0^T f_n^\sigma(t) dt.$$

Each strict integral is one nonnegative term in the assumed cutoff-independent bound. Its limit is finite, and

$$V_n \in L^2(I_*; H^{M+1}), \quad V_{n+1} \in L^2(I_*; H^{M-1}).$$

Summing the finitely many limits gives (C.1.7). The Riesz formula, Sobolev multiplication, and Cauchy–Schwarz in time also give

$$\|Q_n\|_{L^1(I_*; H^M)} \leq C_{M,n} \sum_{a=0}^n \|V_a\|_{L^2(I_*; H^{M+1})} \|V_{n-a}\|_{L^2(I_*; H^{M+1})} < \infty.$$

Thus the pressure rows belong to the terminal record. Dropping higher rows or lowering the spatial height changes no shared derivative or normalized pressure, proving (C.1.8). $\square$

For example, $(N, M) = (2, 3)$ retains V_0, V_1, V_2 in $L^2(I_*; H^4)$, V_1, V_2, V_3 in $L^2(I_*; H^2)$, and Q_0, Q_1, Q_2 in $L^1(I_*; H^3)$. Restriction to $(1, 2)$ drops V_3, Q_2 and reads the remaining coordinates in the corresponding lower spaces. Nothing is recomputed: each retained distribution is the same one. Compatibility concerns this identity; terminal boundedness is the separate quantifier supplied by Assumption 3.4.2.

Each finite coordinate can now be read through its adjacent pair.

Proposition C.1.3 (Adjacent-row recovery). *Under Assumption 3.4.2, every finite n, m has*

$$V_n \in L^2(I_*; H^{m+1}), \quad \partial_t V_n = V_{n+1} \in L^2(I_*; H^{m-1}). \quad (\text{C.1.9})$$

The bound may depend on the chosen pair.

Proof. Choose $(N, M) = (n, m)$. The two terms indexed by n in (C.1.7) are precisely the displayed adjacent rows. They are actual derivatives of u , and restriction preserves that identity. $\square$

The strict and terminal records have always referred to the same derivatives of (u, p) . What has changed is the interval on which their norms are finite; (C.1.8) now makes the terminal family compatible under every decrease of either cutoff.

C.2 The Compatible Intersection

Every finite pair now has a terminal record. Enlarging either cutoff retains the lower record verbatim inside the larger one, because shared entries are the same distributional derivatives of u . Order finite pairs coordinatewise and assemble them with the bonding maps into

$$\mathcal{X}_\infty(I_*) := \varprojlim_{N, M} \mathcal{X}_{N, M}(I_*).$$

A family belongs to this inverse limit precisely when every finite coordinate is an admissible record and

$$\rho_{N', M'; I_*}^{N, M; I_*}(\mathbf{R}_{N, M}) = \mathbf{R}_{N', M'}$$

for every lower pair (N', M') .

Proposition C.2.1 (Compatible projective assembly). *The datum-generated rectangles define*

$$\mathcal{I}[u_0] := (\mathcal{R}_{N, M}[u_0])_{N, M}.$$

This family lies in $\mathcal{X}_\infty(I_)$. Its velocity coordinate is the same distribution u throughout and obeys*

$$u \in \bigcap_{r=0}^{\infty} \bigcap_{m=0}^{\infty} H^r(I_*; H^m(\mathbb{R}^3)). \quad (\text{C.2.1})$$

Coordinate projections commute with all bonding maps.

Proof. Given finite r, m , choose $N \geq r$ and $M \geq m$. That rectangle contains

$$\partial_t^n u \in L^2(I_*; H^{M+1}) \hookrightarrow L^2(I_*; H^m) \quad (0 \leq n \leq r).$$

Because these are the distributional time derivatives of one velocity, $u \in H^r(I_*; H^m)$. The pair was arbitrary, proving (C.2.1). Compatibility makes each lower record the literal projection of every larger one, so no representatives have to be selected or reconciled. $\square$

Project now to the zeroth temporal row while allowing every finite spatial height:

$$\mathcal{S}_0(I_*) := \bigcap_{m=0}^{\infty} L^2(I_*; H^m(\mathbb{R}^3)). \quad (\text{C.2.2})$$

Each $\mathcal{R}_{0,m}[u_0]$ supplies one coordinate, and compatibility identifies them all with the same u . Hence

$$\pi_0(\mathcal{I}[u_0]) = u \in \mathcal{S}_0(I_*).$$

Every finite spatial energy used in the derivative ascent is a coordinate of this row. For $s \geq 0$, set

$$E_s(t) := \frac{1}{2} \|\Lambda^s u(t)\|_2^2. \quad (\text{C.2.3})$$

Choose one integer $m \geq s$. Its coordinate gives

$$\int_0^{T_*} E_s(t) dt \leq C_{s,m} \|u\|_{L^2(I_*; H^m)}^2 < \infty. \quad (\text{C.2.4})$$

Each finite energy is therefore integrable on I_* , with its own bound and the same underlying velocity.

C.3 The Critical Height at the Terminal Face

The terminal rectangles provide adjacent spatial rows. Their first task is to turn two time-integrated norms into one strong value at the terminal face.

Lemma C.3.1 (Adjacent Hilbert-scale trace). *Let $s \in \mathbb{R}$ and suppose*

$$v \in L^2(I_*; H^{s+1}), \quad v_t \in L^2(I_*; H^{s-1}).$$

Then v has one representative in $C([0, T_]; H^s)$, and for $0 \leq a \leq b \leq T_*$,*

$$\left| \|v(b)\|_{H^s}^2 - \|v(a)\|_{H^s}^2 \right| \leq 2 \|v_t\|_{L^2(a,b; H^{s-1})} \|v\|_{L^2(a,b; H^{s+1})}. \quad (\text{C.3.1})$$

Proof. Let $A = (1 - \Delta)^{1/2}$. Time mollification and frequency truncation make the following identity legitimate for smooth approximants:

$$\|v(b)\|_{H^s}^2 - \|v(a)\|_{H^s}^2 = 2 \operatorname{Re} \int_a^b \langle A^{s-1} v_t(t), A^{s+1} v(t) \rangle_{L^2} dt. \quad (\text{C.3.2})$$

Cauchy–Schwarz gives (C.3.1). The two assumed L^2 rows pass the right-hand side to the limit. The limit identity makes the H^s norm continuous, while the approximation also gives weak H^s continuity; together they imply strong continuity. Any two continuous representatives agree everywhere because they agree almost everywhere. $\square$

The critical readout needs only $\mathcal{R}_{0,2}[u_0]$. Its integer rows embed into the fractional adjacent pair

$$u \in L^2(I_*; H^{5/2}), \quad u_t \in L^2(I_*; H^{1/2}). \quad (\text{C.3.3})$$

No fractional rectangle has been assumed; these are continuous embeddings of the integer record. The trace lemma at $s = 3/2$ now gives

$$u \in C([0, T_*]; H^{3/2}). \quad (\text{C.3.4})$$

Let $H = E_{1/2}$. To read the homogeneous critical energies from the inhomogeneous trace, mollify in time and truncate frequencies to $\varepsilon \leq |\xi| \leq R$. Differentiation and Plancherel then give

$$\frac{1}{2} \left\| \Lambda^{3/2} u(b) \right\|_{L^2}^2 - \frac{1}{2} \left\| \Lambda^{3/2} u(a) \right\|_{L^2}^2 = \operatorname{Re} \int_a^b \left\langle \Lambda^{1/2} u_t(t), \Lambda^{5/2} u(t) \right\rangle_{L^2} dt.$$

Cauchy–Schwarz passes the integral to the limit through the two critical rows; the continuous $H^{3/2}$ trace passes the endpoint terms. Removing all cutoffs yields, for $0 \leq a \leq b \leq T_*$,

$$E_{3/2}(b) - E_{3/2}(a) = \operatorname{Re} \int_a^b \left\langle \Lambda^{1/2} u_t(t), \Lambda^{5/2} u(t) \right\rangle_{L^2} dt.$$

The adjacent pair makes the derivative integrable, and the complete spatial row makes $E_{3/2}$ itself integrable. Hence

$$E_{3/2} \in W^{1,1}(I_*) \cap L^1(I_*) \subset L^\infty(I_*). \quad (\text{C.3.5})$$

The integer rows also put the derivative of the critical height in L^1 :

$$H'(t) = \left\langle \Lambda^{1/2} u(t), \Lambda^{1/2} u_t(t) \right\rangle_{L^2} \in L^1(I_*). \quad (\text{C.3.6})$$

Keep the pressure-explicit production current as

$$\mathcal{P}_{1/2} := -\operatorname{Re} \left\langle \Lambda^{1/2} u, \Lambda^{1/2} \left((u \cdot \nabla) u + \nabla p \right) \right\rangle_{L^2}.$$

Pairing the full equation with Λu gives

$$H' = -2\nu E_{3/2} + \mathcal{P}_{1/2}.$$

The two integrable rows consequently imply

$$H \in W^{1,1}(I_*), \quad E_{3/2} \in L^\infty(I_*) \cap L^1(I_*), \quad \mathcal{P}_{1/2} \in L^1(I_*).$$

The height and $E_{3/2}$ therefore reach the terminal face continuously; the production current is only integrable, and no endpoint value is claimed for it. This entire critical trace comes from $\mathcal{R}_{0,2}[u_0]$. Higher rectangles will repeat the endpoint argument at every finite Sobolev height.

C.4 One Endpoint for the Whole Tower

A restart needs an actual datum at T_* , not a separate endpoint in each rectangle. Under Assumption 3.4.2, Proposition C.1.2 supplies the complete spatial row on a hypothetical finite branch. The equation then supplies enough time integrability to make the same velocity converge at every finite height.

Lemma C.4.1 (The complete spatial row reaches every endpoint topology). *Let (u, p) be the fixed normalized history on I_* . If $u \in \mathcal{S}_0(I_*)$, then for every $m \in \mathbb{N}_0$,*

$$u_t \in L^1(I_*; H^m), \quad u \in W^{1,1}(I_*; H^m) \hookrightarrow C([0, T_*]; H^m). \quad (\text{C.4.1})$$

Proof. The pressure-complete transport response can be written without separating pressure:

$$(u \cdot \nabla)u + \nabla p = \mathbb{P} \nabla \cdot (u \otimes u).$$

Under the fixed normalization, this is exactly transport plus ∇p . Sobolev multiplication and boundedness of the Leray and Riesz multipliers give

$$\|\mathbb{P} \nabla \cdot (u \otimes u)\|_{H^m} \leq C_m \|u\|_{H^{m+2}}^2.$$

The complete row, together with $T_* < \infty$, therefore yields

$$\begin{aligned} \|\mathbb{P} \nabla \cdot (u \otimes u)\|_{L^1(I_*; H^m)} &\leq C_m \|u\|_{L^2(I_*; H^{m+2})}^2 < \infty, \\ \|\Delta u\|_{L^1(I_*; H^m)} &\leq T_*^{1/2} \|u\|_{L^2(I_*; H^{m+2})} < \infty. \end{aligned}$$

The momentum equation puts u_t in $L^1(I_*; H^m)$. In particular, for $0 \leq s < t < T_*$,

$$\|u(t) - u(s)\|_{H^m} \leq \nu(t-s)^{1/2} \|u\|_{L^2((s,t); H^{m+2})} + C_m \|u\|_{L^2((s,t); H^{m+2})}^2.$$

Both terms vanish when $s, t \uparrow T_*$, making the same $u(t)$ Cauchy in H^m . Since the finite interval also embeds $L_t^2 H_x^m$ into $L_t^1 H_x^m$, $u \in W^{1,1}(I_*; H^m)$; the Bochner fundamental theorem of calculus supplies the continuous representative and its endpoint. $\square$

Let $u_*^{(m)}$ be the endpoint found in H^m . If $k > m$, both the H^k trace embedded into H^m and the native H^m trace represent the same u almost everywhere. Continuity makes them identical through the closed interval. Their terminal values are therefore one element

$$u_* := u(T_*) \in \bigcap_{m=0}^{\infty} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3). \quad (\text{C.4.2})$$

No uniformity in m is used. Divergence passes continuously to the limit, so u_* is solenoidal, and the strong L^2 trace preserves finite kinetic energy.

Apply the same adjacent trace to every temporal coordinate:

$$\partial_t^N u \in C([0, T_*]; H^m) \quad (N, m \in \mathbb{N}_0).$$

For fixed N , spatial compatibility identifies all these traces as

$$V_N^* := \lim_{t \uparrow T_*} \partial_t^N u(t) \in H^\infty(\mathbb{R}^3). \quad (\text{C.4.3})$$

It remains to show that this trace family is the actual Navier–Stokes jet generated by u_* .

Proposition C.4.2 (Pressure-complete endpoint jet). *The endpoint u_* determines one sequence $(V_N^*, Q_N^*)_{N \geq 0} \subset H^\infty \times H^\infty$ by*

$$\begin{aligned} V_0^* &= u_*, \\ Q_N^* &= R_i R_j \sum_{a=0}^N \binom{N}{a} V_{a,i}^* V_{N-a,j}^*, \end{aligned} \quad (\text{C.4.4})$$

$$V_{N+1}^* = \nu \Delta V_N^* - \sum_{a=0}^N \binom{N}{a} (V_a^* \cdot \nabla) V_{N-a}^* - \nabla Q_N^*. \quad (\text{C.4.5})$$

This sequence is exactly the family of left traces of the original normalized velocity and pressure history.

Proof. The space $H^\infty = \bigcap_m H^m$ is stable under multiplication and spatial differentiation, and Riesz transforms preserve every one of its Sobolev coordinates. The two recurrences therefore determine a unique next pair at each finite order.

Fix a finite target height m and take the trace convergence in a Sobolev space high enough for every product and derivative in the recurrence. All operations are continuous between these finite spaces, so the limit of (C.1.3)–(C.1.4) is the displayed endpoint recurrence. Starting from $V_0^* = u_*$, induction in N identifies the recursively generated entries with the left traces. Since m was arbitrary, the identity holds in H^∞ , and spatial differentiation supplies every mixed endpoint coordinate. $\square$

No new pressure constant appears at the terminal face. The decaying Riesz formula fixes every $Q_N(t)$ on the left, and the limit preserves that normalization. Thus inserting u_* into the normalized recurrence produces exactly the left mixed jet.

C.5 Restarting the Same History

Use the endpoint itself as the restart datum. The local theorem gives a right-hand solution on one interval for all finite Sobolev heights. Its jet and the left trace jet both begin from u_* and obey the same normalized recurrence; the matching jets make the join smooth, and uniqueness makes the two pieces one history.

Lemma C.5.1 (Smooth local restart and uniqueness). *Let $t_0 \geq 0$ and $a \in H_\sigma^\infty(\mathbb{R}^3)$. There are $\delta > 0$ and a unique smooth normalized pair (w, q) on $[t_0, t_0 + \delta] \times \mathbb{R}^3$, with $w(t_0) = a$ and*

$$q = R_i R_j (w_i w_j).$$

All finite spatial Sobolev orders persist on this same interval. The temporal jet at t_0 is generated by (C.4.4)–(C.4.5).

Proof. Translate time and apply Proposition A.1.1 to a . Its lifespan depends only on ν and $\|a\|_{H^3}$, while persistence carries every finite height of a on that interval. The normalized pair is unique. Differentiating at the initial face and using the Riesz pressure formula gives the endpoint recurrences. $\square$

Theorem C.5.2 (Global continuation from whole-terminal rectangles). *Let u_0 and all its spatial derivatives be rapidly decaying, with $\nabla \cdot u_0 = 0$, and fix $\nu > 0$. Let (u, p) be its maximal normalized classical history. If a hypothetical finite branch satisfies Assumption 3.4.2, then $T_* = \infty$ and (u, p) is smooth on every finite interval.*

Proof. Suppose instead that $T_* < \infty$. The hypothesis supplies every terminal rectangle with compatible overlaps. Their intersection produces the common endpoint $u_* \in H_\sigma^\infty$ and its actual normalized jet. Restart the local theorem at T_* from u_* , obtaining (w, q) on $[T_*, T_* + \delta]$, and set

$$(\tilde{u}, \tilde{p})(t) := \begin{cases} (u, p)(t), & 0 \leq t < T_*, \\ (w, q)(t), & T_* \leq t \leq T_* + \delta. \end{cases}$$

The two velocities meet at u_* . Every one-sided pressure and velocity derivative also agrees because both jets are generated from that datum by the same normalized recurrences. The piecewise pair is smooth across T_* and solves the same fixed-viscosity equation beyond it. Strong uniqueness makes it the continuation of the original datum-generated history, contradicting maximality. Hence

$$T_* = \infty.$$

On each finite interval, every finite temporal and spatial coordinate remains in the compatible intersection. Sobolev embedding then gives

$$u, p \in C^\infty([0, \infty) \times \mathbb{R}^3).$$

$\square$

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